

Taiwan Explorers Planning Report (M1)

GROUP 18 FLOWERS

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GITHUB REPOSITORY LINK

<https://github.com/CMPT-276-SUMMER-2025/final-project-18-flowers>

Project Planning

1. Finalized APIs

API 1: Google Maps API

Link: [Google Maps Platform Documentation](#) | [Google for Developers](#)

The Google Maps API allows us to integrate a variety of features and data from Google Maps into our web application. It enables the inclusion of interactive maps customized with content, location-based services, and access to Google's extensive global dataset of places. We plan to leverage this by retrieving images of locations (e.g., attractions, cities), place descriptions, and specific details such as reviews and costs. In our application, we'll use an interactive map focused on various regions of Taiwan (e.g., Taipei, Hualien, Yilan), with markers that display relevant information. Additionally, we intend to use static maps on the attractions pages (e.g., Taipei 101) to provide users with a basic sense of each location's geography. Route maps with multiple transportation modes will be integrated into the itinerary generator to help users plan their travel efficiently.

Feature 1: Integrate Google's Vast Database on Places

Display location and information data about various attractions, accommodations, and more using Google Maps API's Places API.

- **Benefits:**
 - Users will be able to see what places look like and gain a first impression of the place to consider for their trip.
 - Users will be able to find accommodations near places to find hotels, restaurants, or stores.
 - Users will be able to get information such as a description, real reviews, and cost to be able to make smart decisions and make personal fulfilling plans.

(Changed Feature 1's title and description to focus more on Google Places API and its location-based data, rather than general interactive map features.)

Feature 2: Interactive Dynamic Maps

Display interactive maps of various locations or routes that include markers with popups, custom styling, and customizability of data displayed on the map.

- **Benefits:**

- Users will be able to click on markers to get helpful information immediately through a pop-up.
- Users will be able to filter or customize the map to fit their needs, such as finding a specific type of location.
- Users will be able to visually explore different locations and routes in an engaging way.

(Added new and updated current benefits to feature 2. Furthermore, the title of Feature 2 and its description were changed from a static map to an interactive map.)

Feature 3: Return Route Data and Display It

Calculate and display routes between a series of markers and return specific directions for transportation such as transit, driving, walking, and more.

- **Benefits:**

- Users will be able to see visually how long it'll take to traverse between various locations and fit it into their travel plan.
- Users will be able to compare how long a route is based on the mode of transportation.
- Users will be able to get a gauge of the total time.

(Replaced the original street view feature with a more user-relevant feature of route calculation and directions. Since the original feature has been replaced, an updated description for the new feature is now available.)

API 2: Google Gemini API

Link: <https://ai.google.dev/gemini-api/docs>

The Google Gemini API allows us to integrate Google's advanced AI model, Gemini, into our web application to generate dynamic and personalized content. This enables features such as AI-powered chatbots and content summarization. One planned feature is to generate AI-based summaries for location markers on the map, covering places like Taipei or Hualien, and providing users with quick and informative overviews. For the itinerary feature, we aim to process user-provided travel preferences and convert them into natural language prompts for Gemini. The model will then generate a custom itinerary that follows a specific template. Additionally, we plan to implement an AI chatbot that responds to user messages using keyword

detection or preset selectable options. These options may include sub-options to help users find relevant information or navigate to different pages.

Feature 1: Itinerary Generator

AI trip planner with Google Services Integrations by utilizing Google Gemini API and Google Maps API. Users can personalize their itinerary with factors such as time, cost, location, and interests.

- **Benefits:**
 - Users will be able to receive a personalized travel plan based on their schedule, budget, and preferences.
 - Users will be able to save time by having an AI automatically generate optimized routes based on the users' input.
 - Users will be able to make more informed travel decisions with up-to-date costs and time considerations.

(Simplified the original title from Smart Trip Planner to Itinerary Generator. Furthermore, the original description has been expanded to clarify that the itinerary generator uses both the Google Gemini API and the Google Maps API.)

Feature 2: AI Bot

A chatbot and FAQ assistant powered by the Google Gemini API that provides real-time answers to users' questions. The bot will deliver fast responses in a conversational format to any user's inquiries.

- **Benefits:**
 - Users will be able to quickly get answers to commonly asked questions without needing to search through the site manually.
 - Users will be able to receive assistance at any time, ensuring timely responses whenever needed.
 - Users will be able to save time by receiving answers that are specific to their question.

(Rewrote and improved the wording of the original title, as well as added a new and much clearer description of what the feature does.)

Feature 3: AI Summary of Places

When a user clicks on a location marker, such as Jiufen on the map, the Gemini AI will generate and display a short and descriptive summary of the location, for example:

“Jiufen is a historic mountain town known for tea houses and resemblances to scenes from the film Spirited Away”.

- **Benefits:**
 - Users will be able to instantly learn about a location's information without leaving the map or conducting external research.
 - Users will be able to explore places more efficiently by seeing concise descriptions that highlight key features.
 - Users will be able to gain a better understanding of each location's cultural or historical context through brief explanations.

(Replaced the original budget-based feature (merged into the first feature) with a more location-based summary feature, alongside a rewritten description of the new feature.)

2. Mid-fidelity Wireframe (Figma)

- **Figma Link:**
<https://www.figma.com/design/ewFkHSWJKsQiulEi9ak8Ed/Taiwan-Explorers-Wireframing?node-id=161-109&t=LYiCb9il8Ppn9lsx-1>

Mid-fidelity Wireframe Screenshots:

3. SDLC Model

- **Model:** Agile Kanban
- **Explanation of why we chose Kanban SDLC Model**

Despite the numerous other SDLC models that we could have chosen, our group decided on following the Agile Kanban model. Other SDLC models were not as appealing to us and were harder to interpret. However, the Agile model caught our attention for being the most popular SDLC model as well as having attributes such as flexibility, roles-based, cyclical, and collaborative. The collaborative aspect of Agile piqued our interest since this is a group project requiring multiple people of differing programming proficiencies to work together. Furthermore, the roles-based nature of this model was enticing because the members of our group have different strengths and weaknesses which can be addressed via assigning roles to leverage our individual expertise. Unlike other SDLC, the Agile model is better at adapting to change which we felt was important to us as a team with varying degrees of experience. This trait is

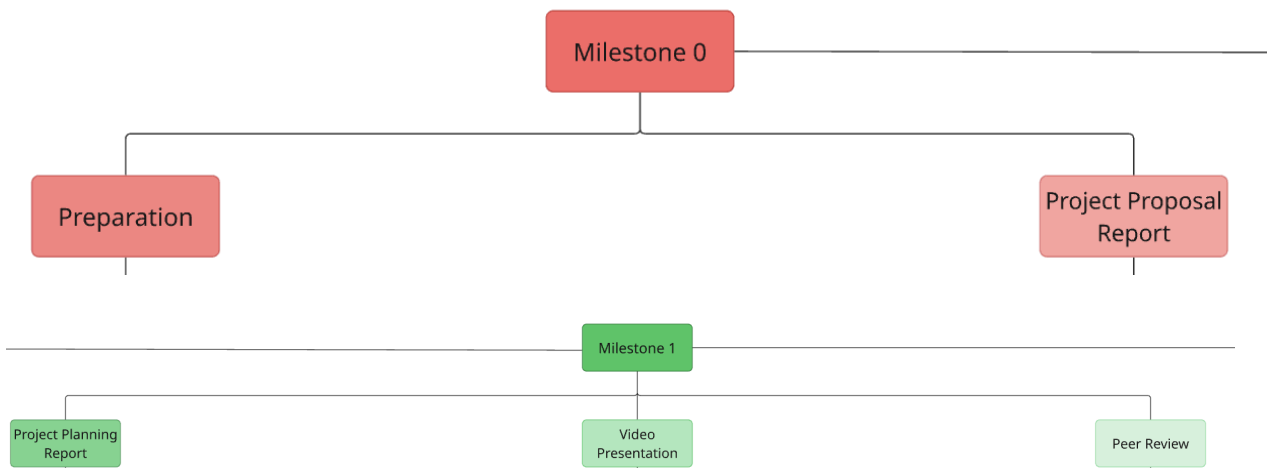
facilitated by standups and retrospective meetings where we can discuss our personal progress as well as how the overall project is progressing and if any changes need to be made. In addition, our group was intrigued by the Agile manifesto. The point, “responding to change over following a plan” is one that we will apply to this project. Finally, we chose Kanban as the Agile framework as it included the use of “cards” (issues in the context of GitHub) and “lanes” to track the progression of various tasks pertaining to the project. We felt that this was the best way to get a sense of how the project is coming along.

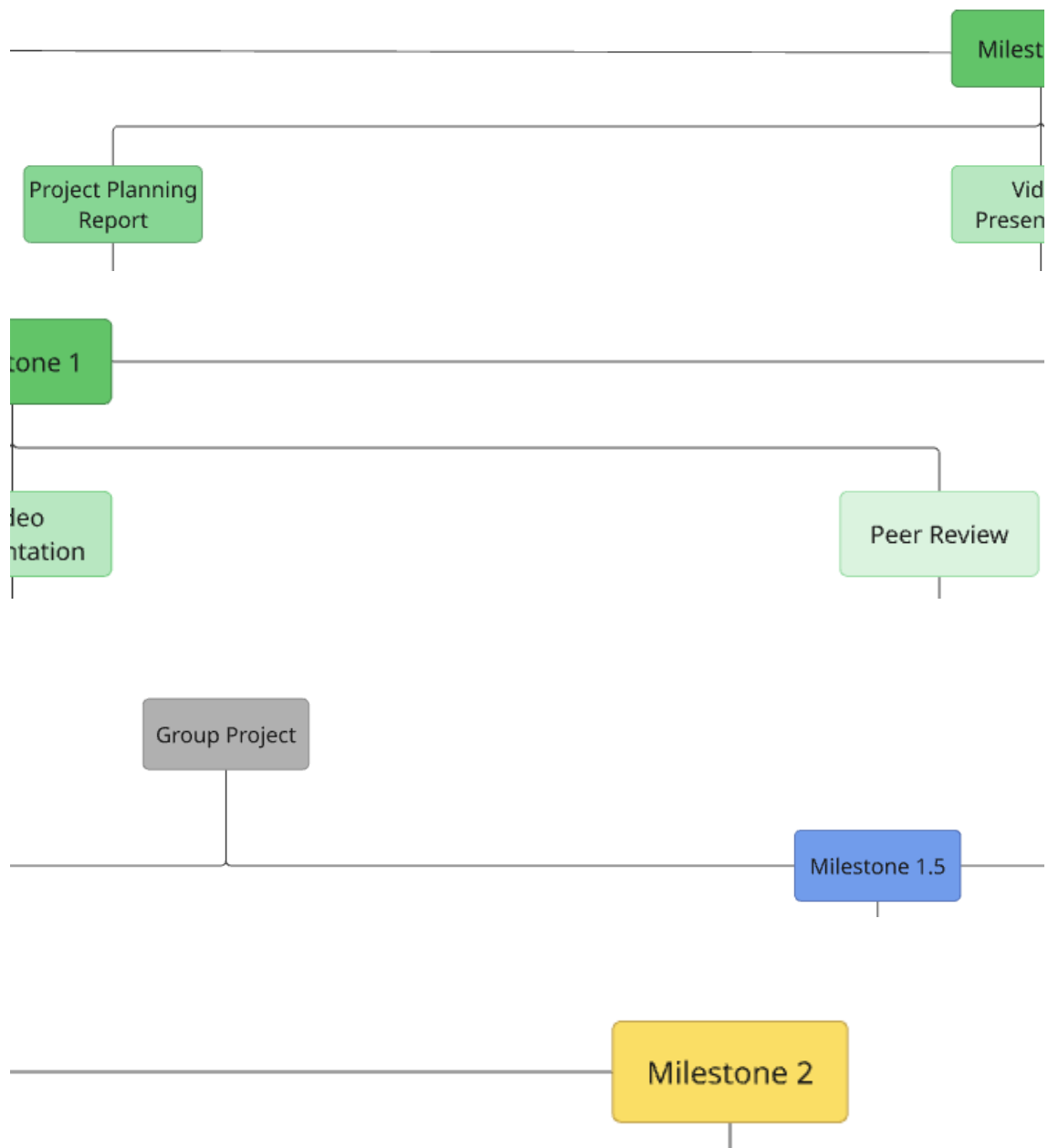
5. Work Breakdown Structure (Miro)

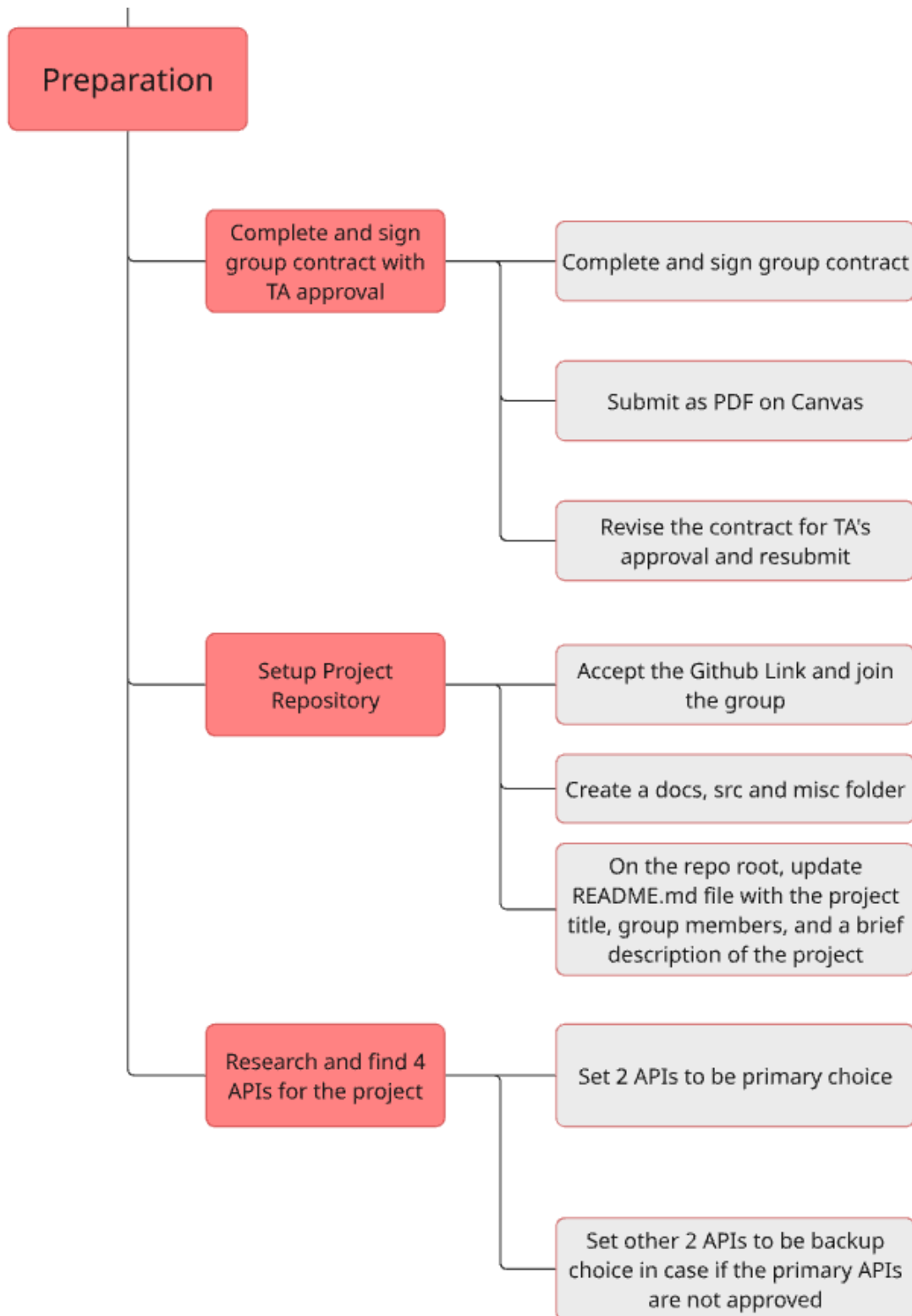
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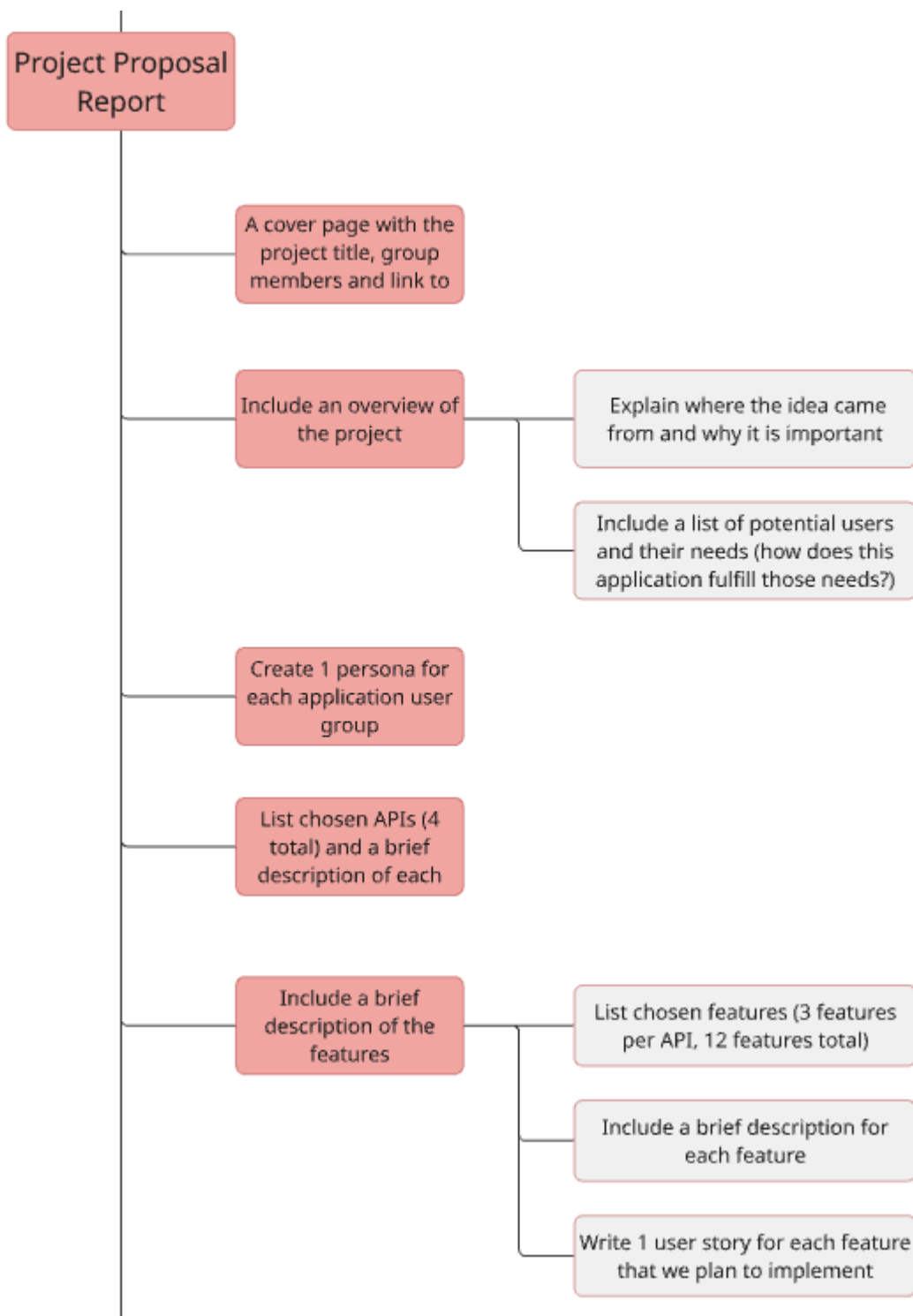
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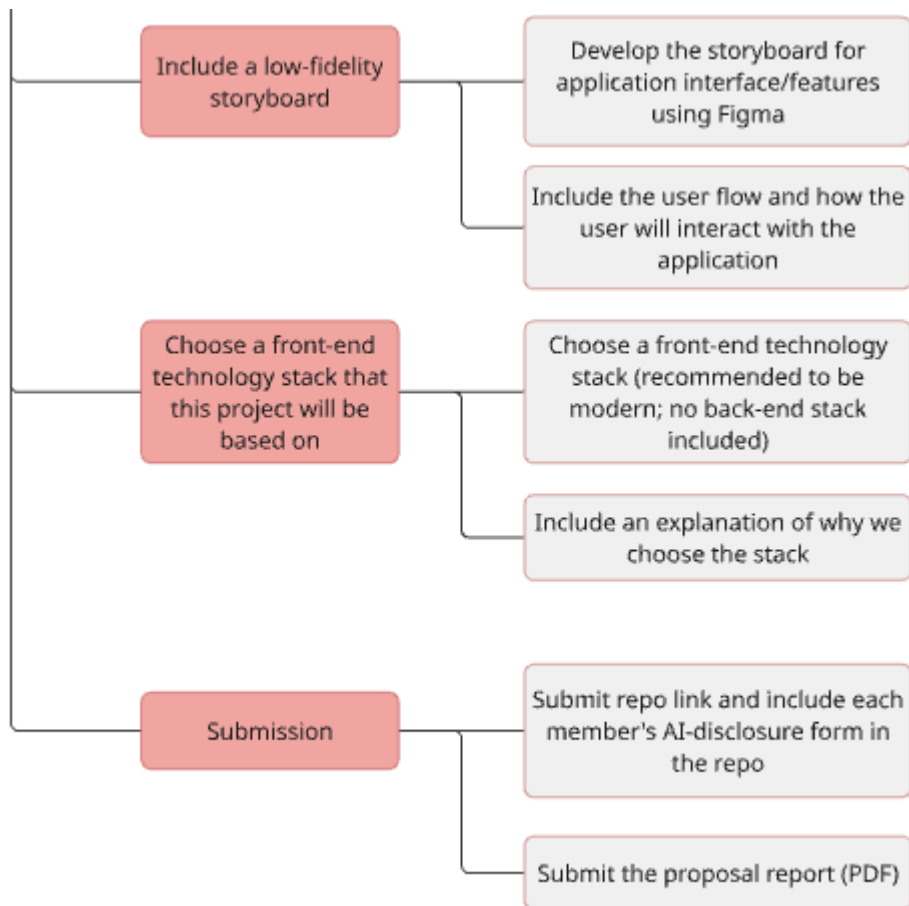
Work Breakdown Structure Screenshot:

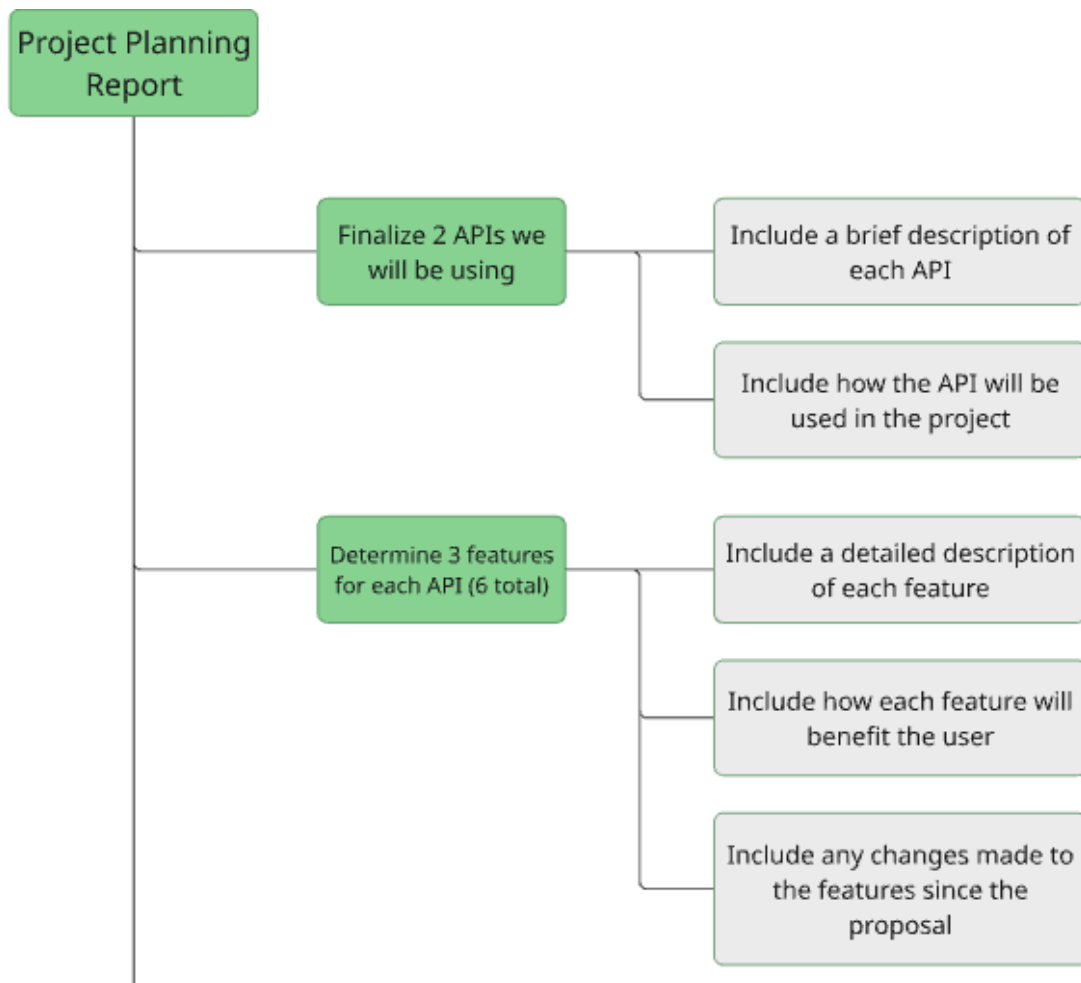


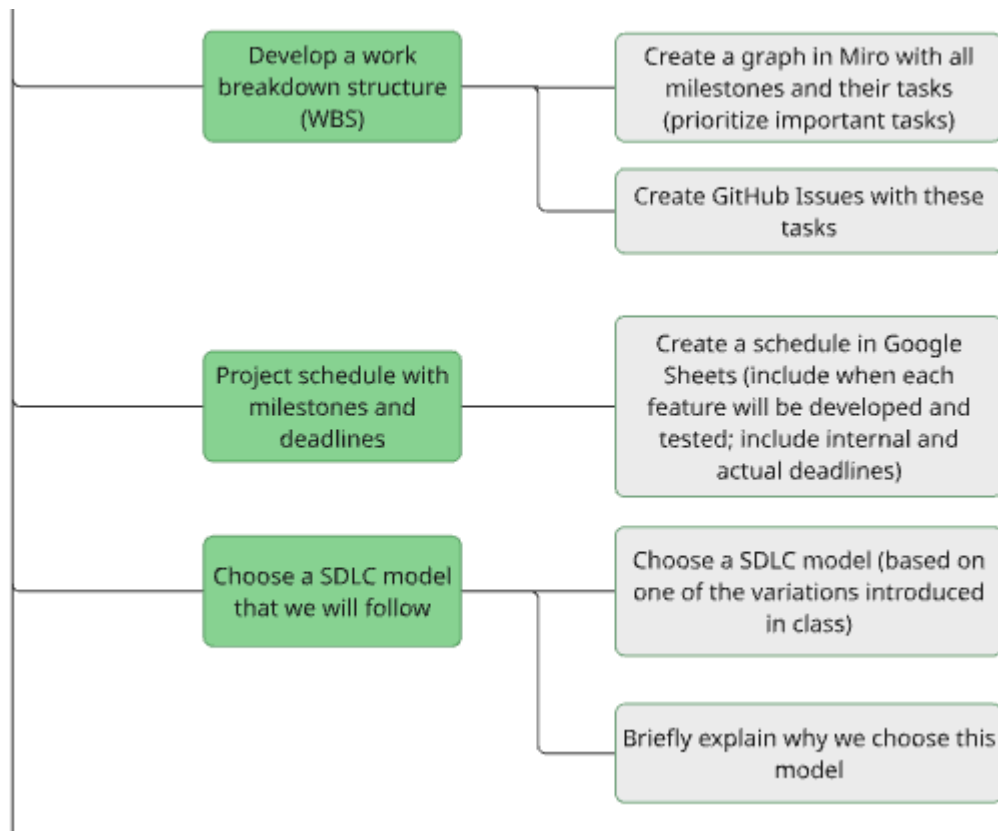


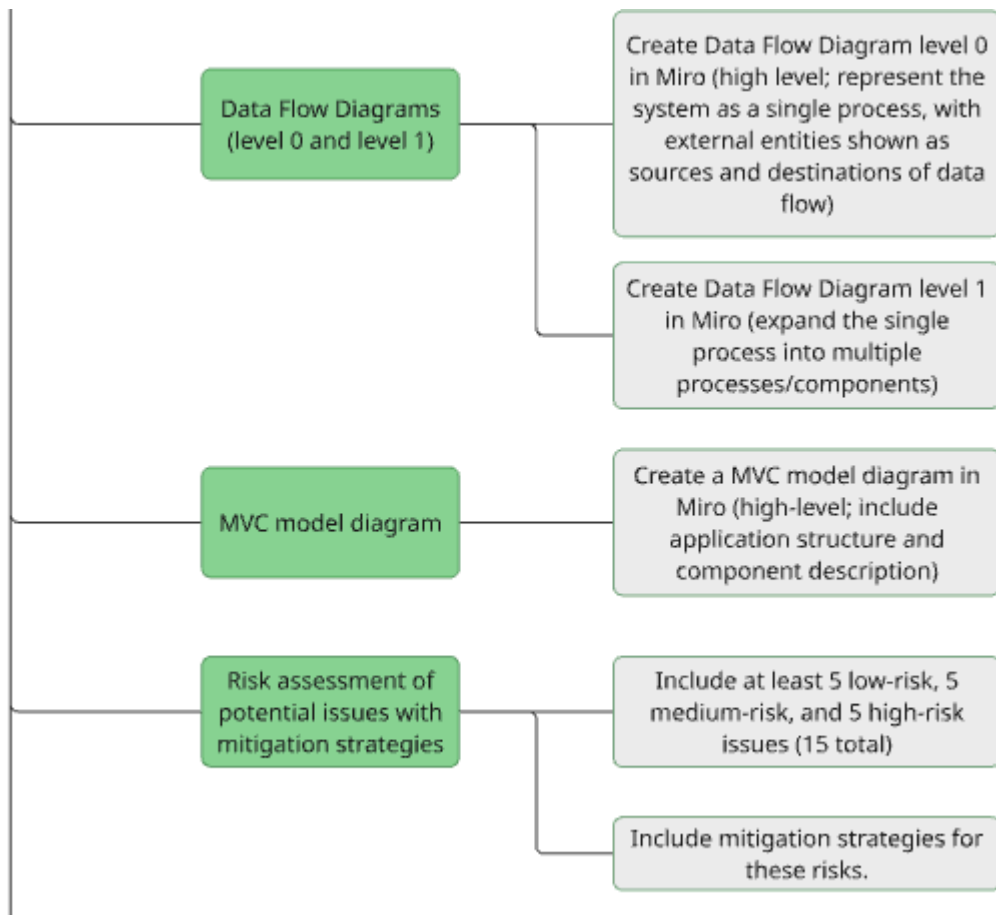


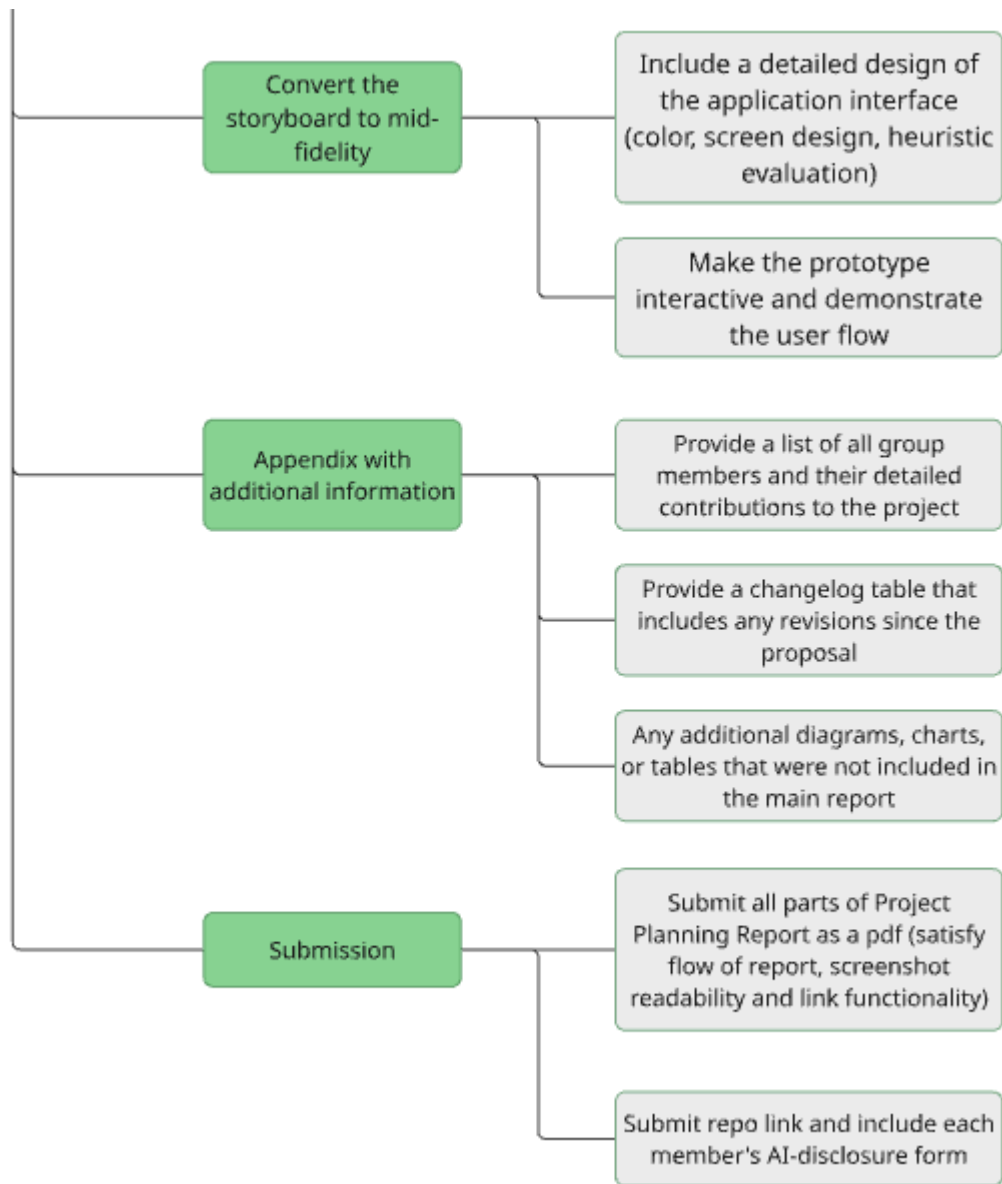


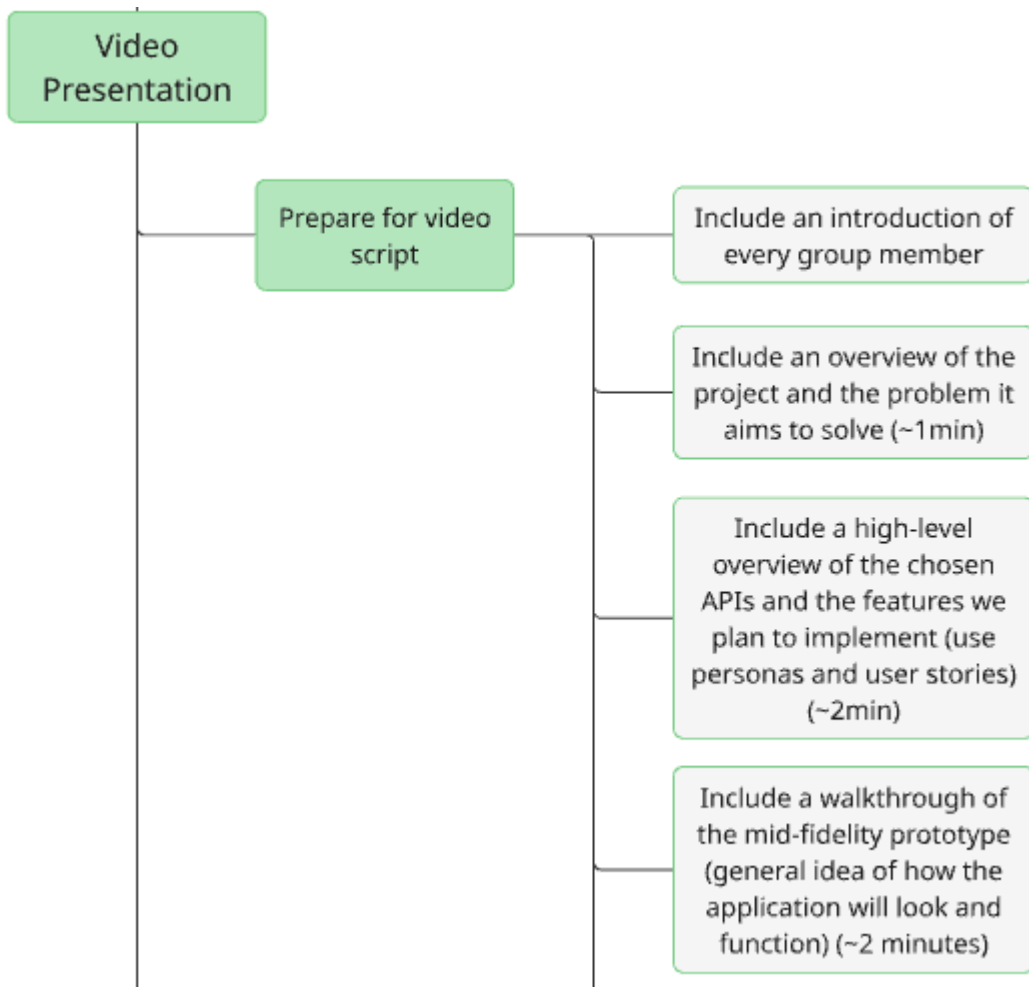


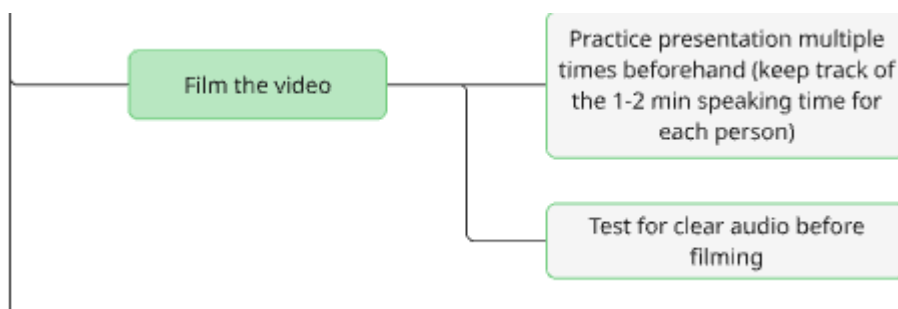
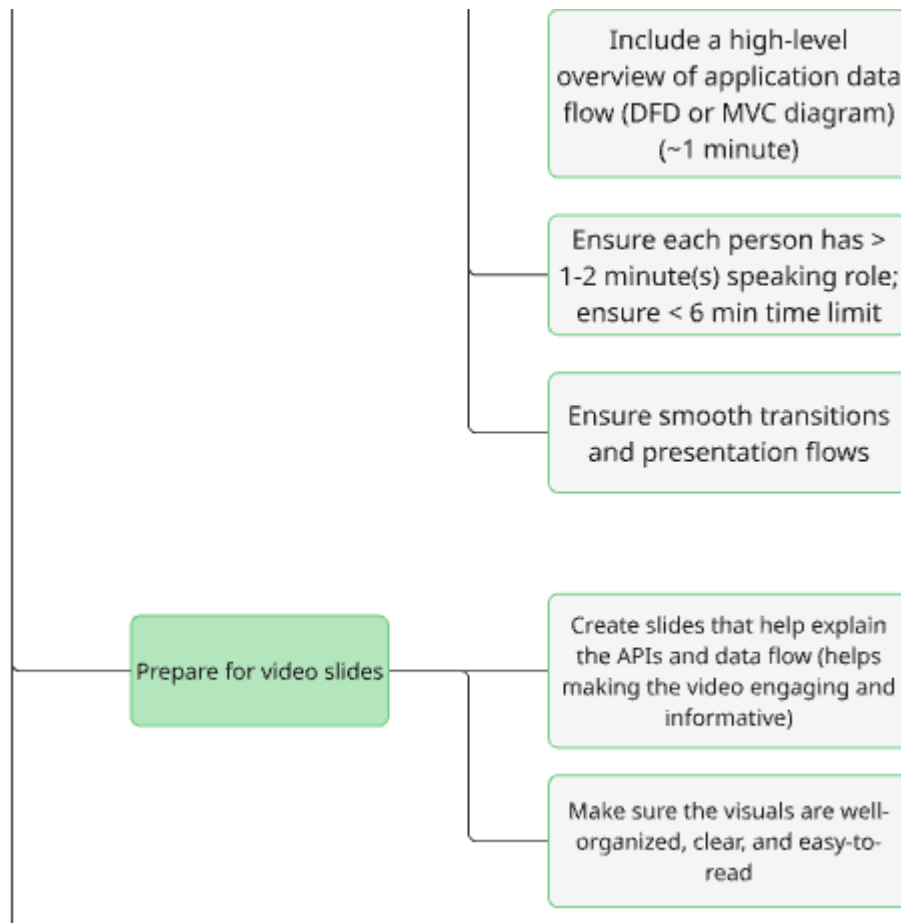


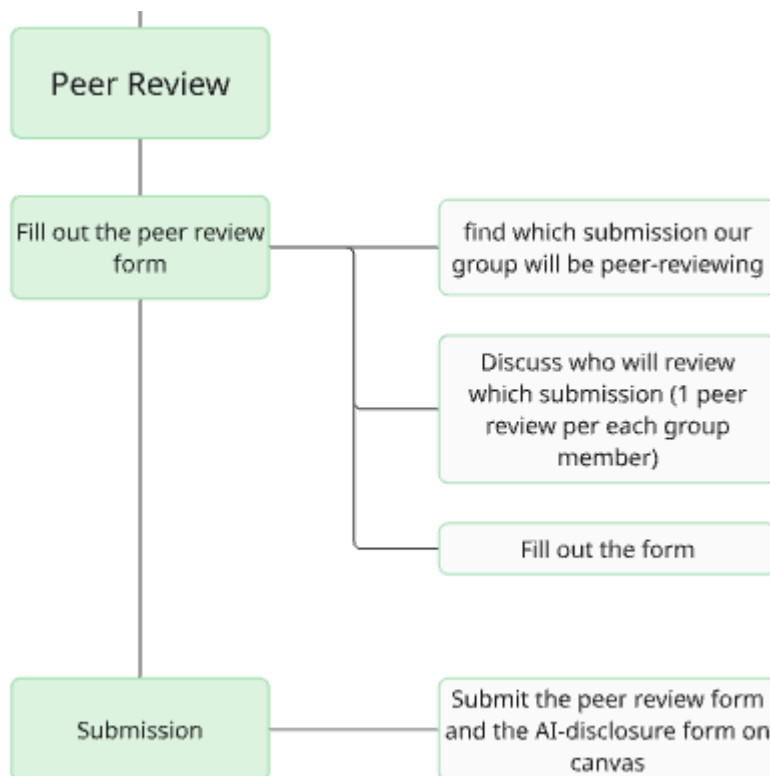
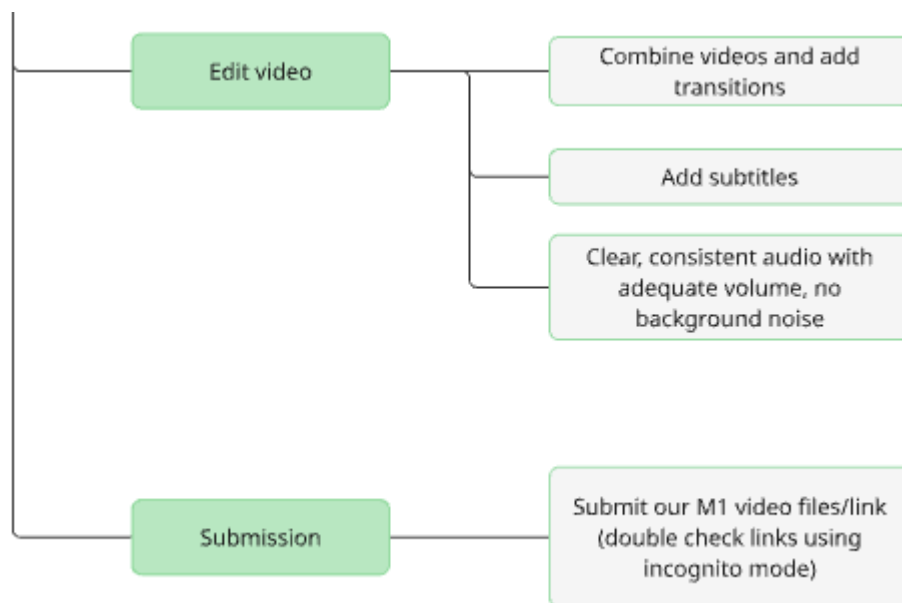


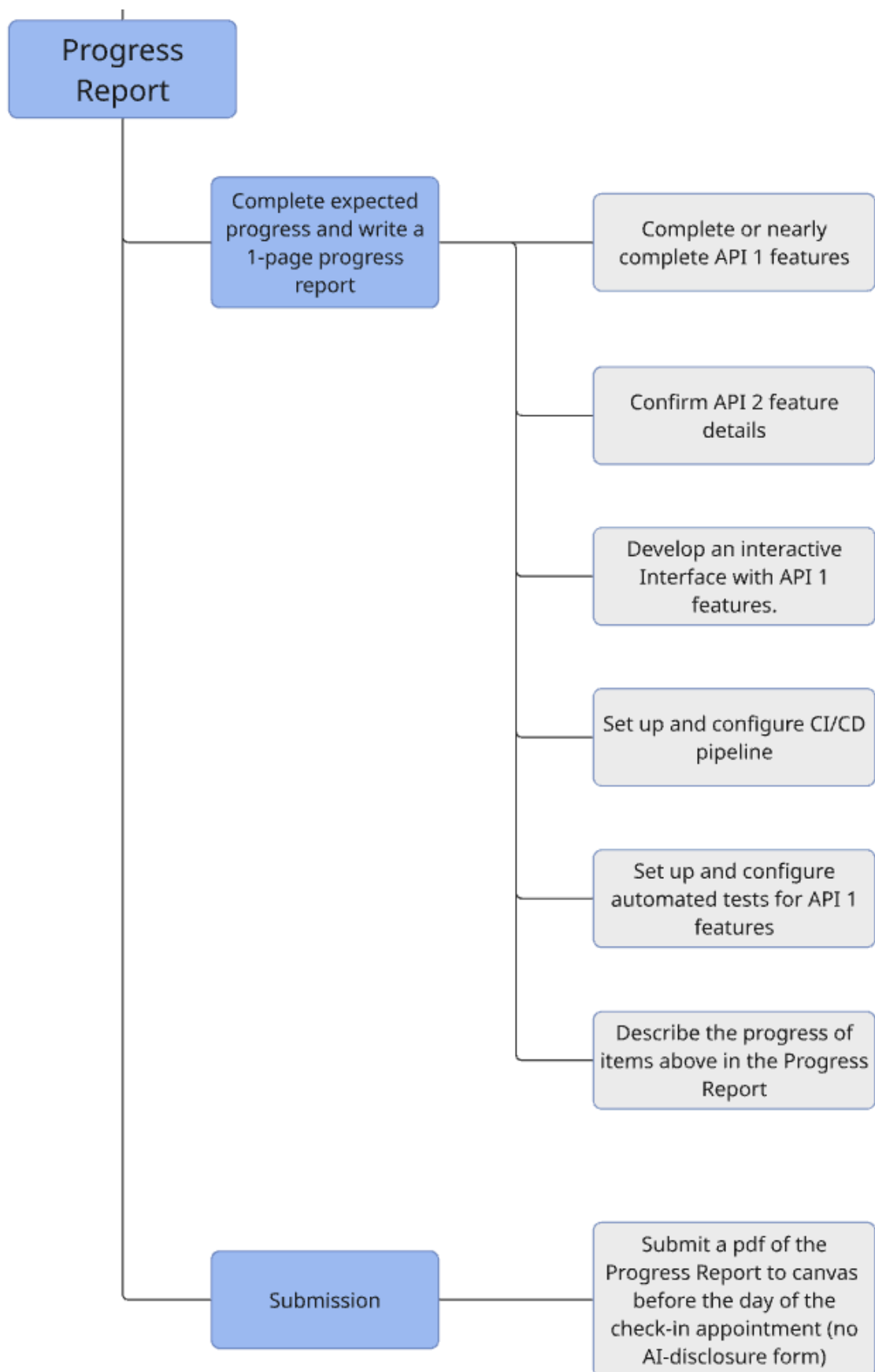


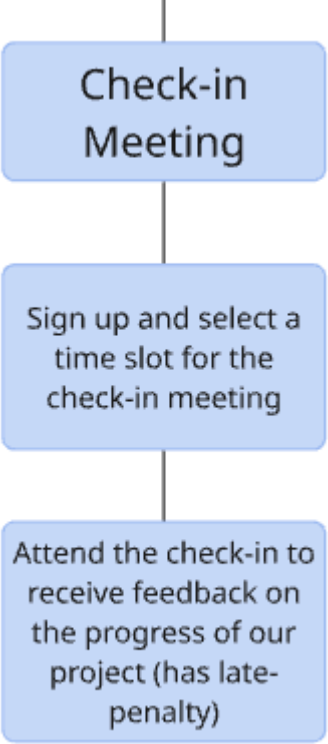










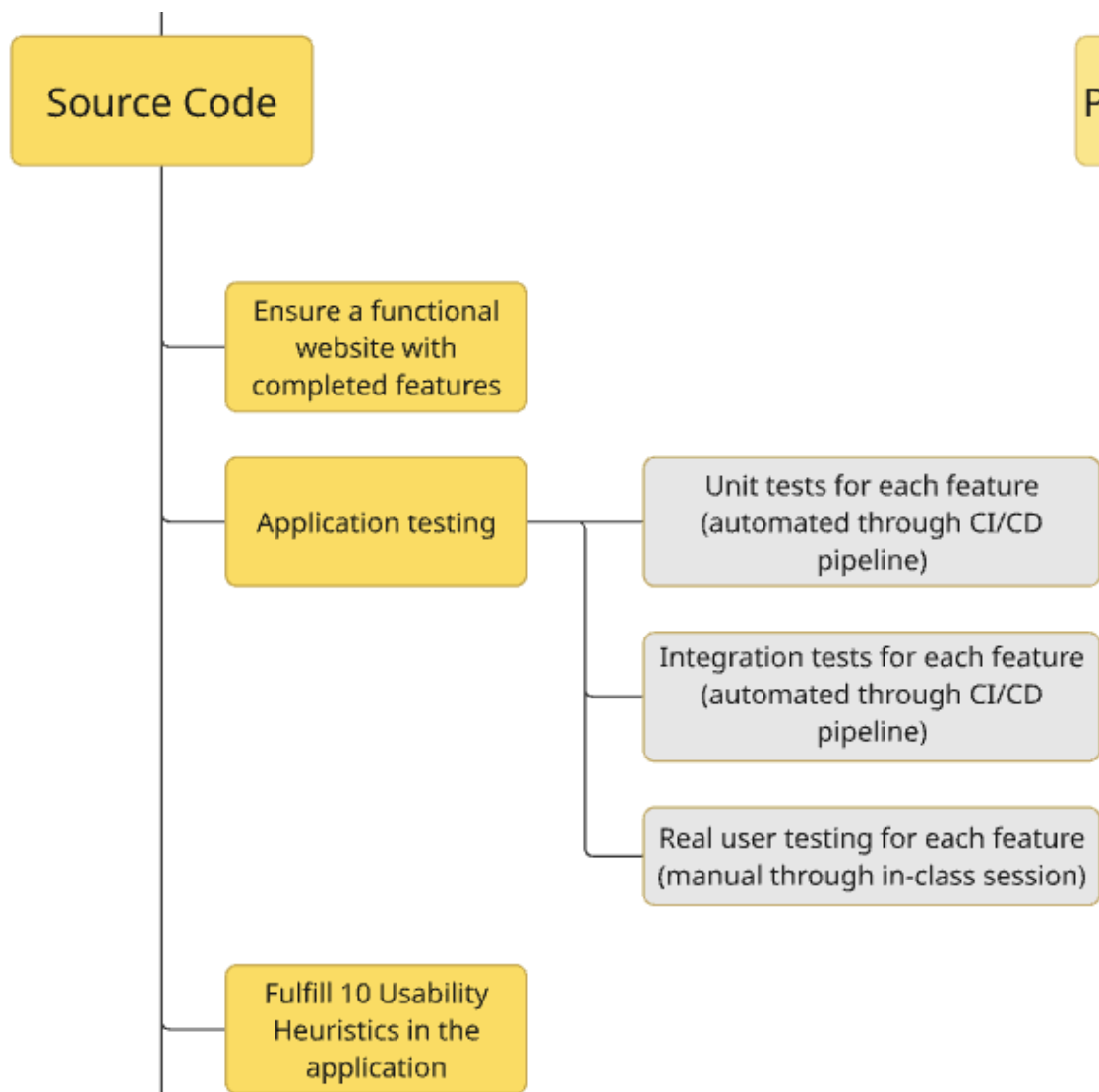


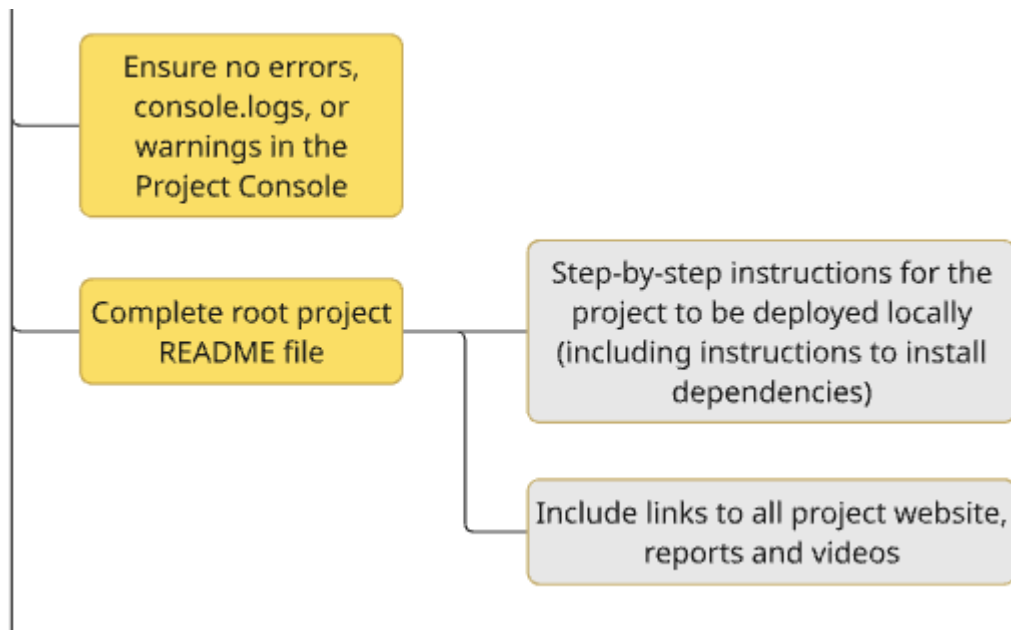
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graph TD; A[Check-in Meeting] --> B[Sign up and select a time slot for the check-in meeting]; B --> C[Attend the check-in to receive feedback on the progress of our project (has late-penalty)];
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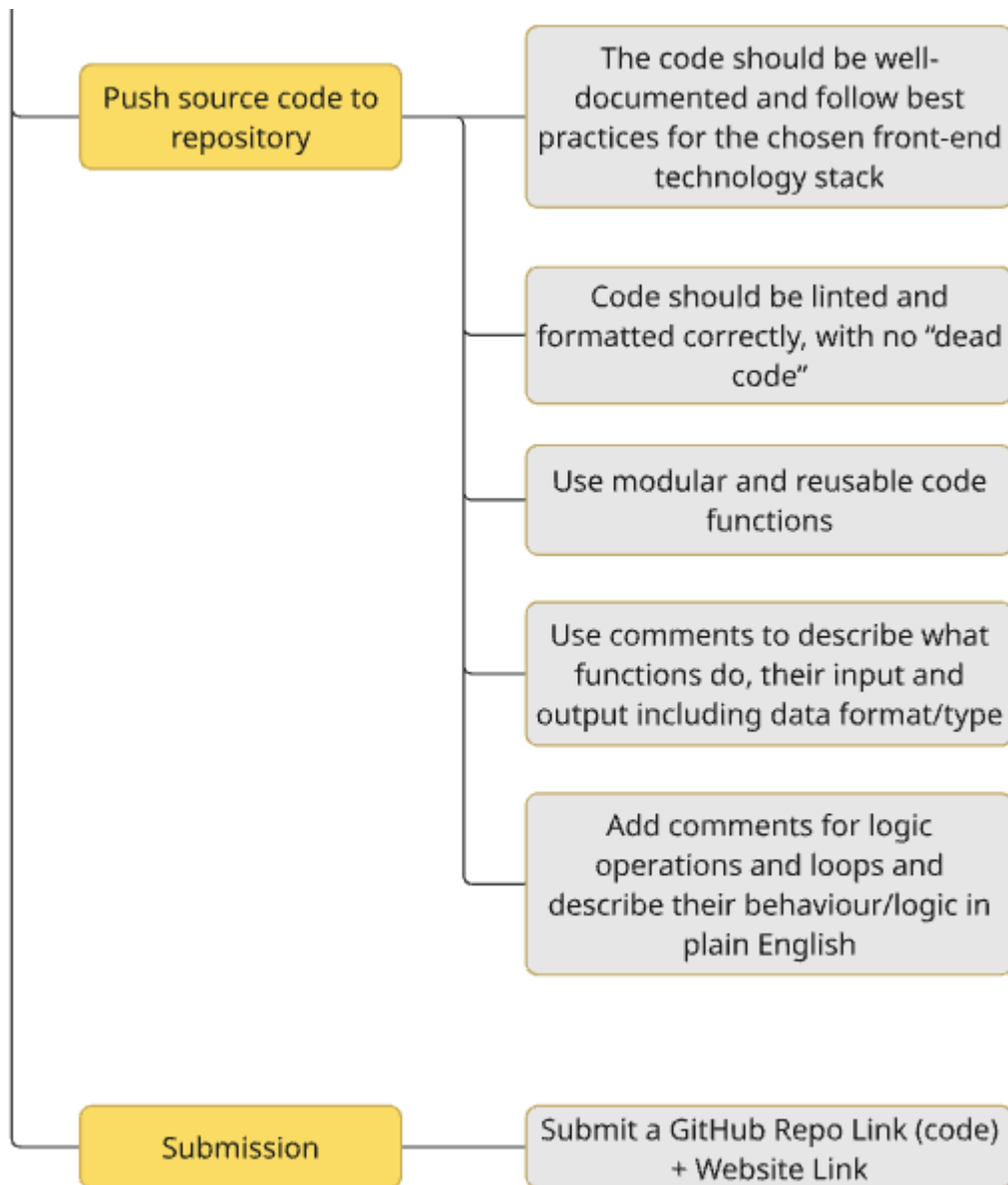
Check-in Meeting

Sign up and select a time slot for the check-in meeting

Attend the check-in to receive feedback on the progress of our project (has late-penalty)



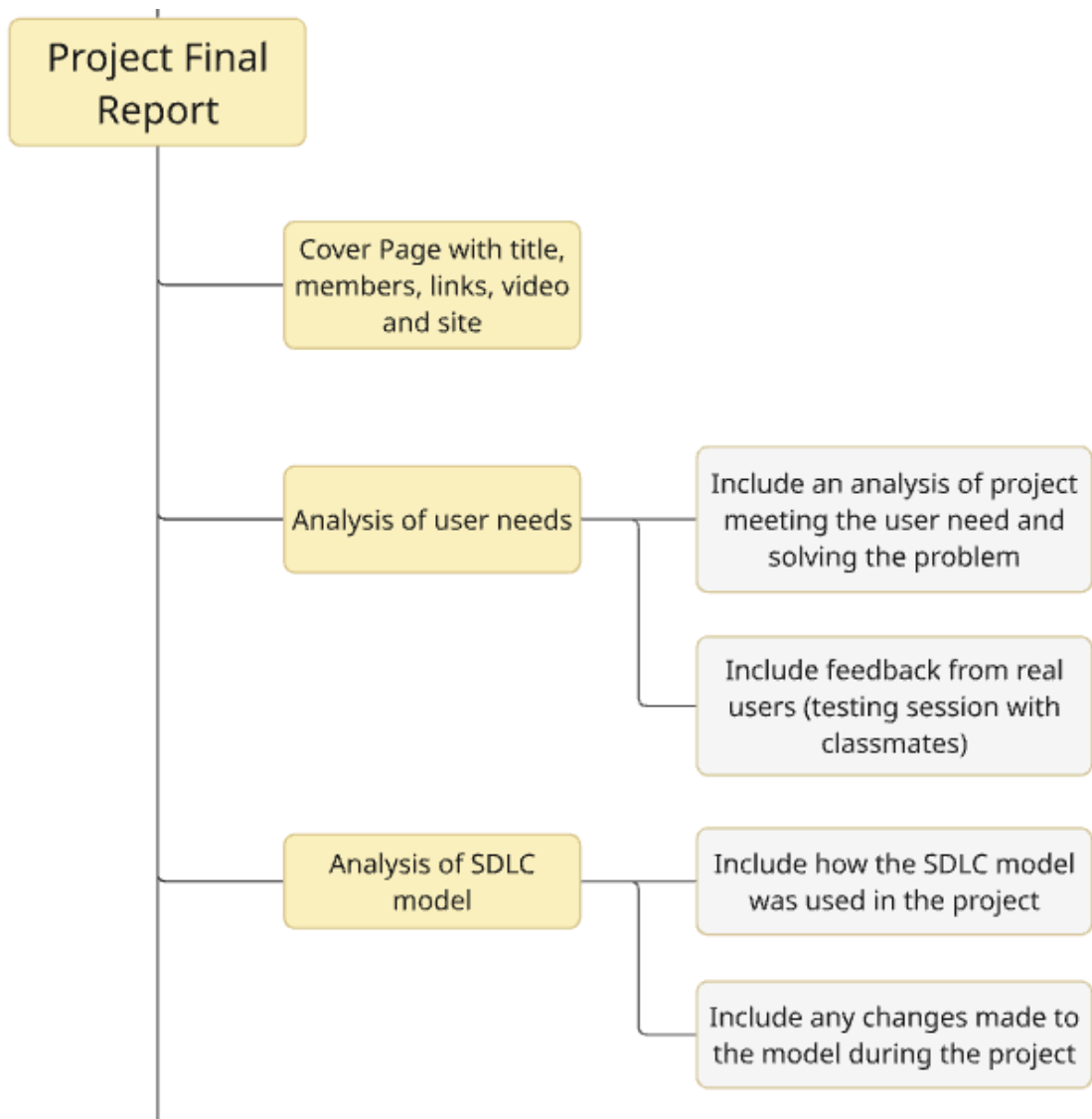


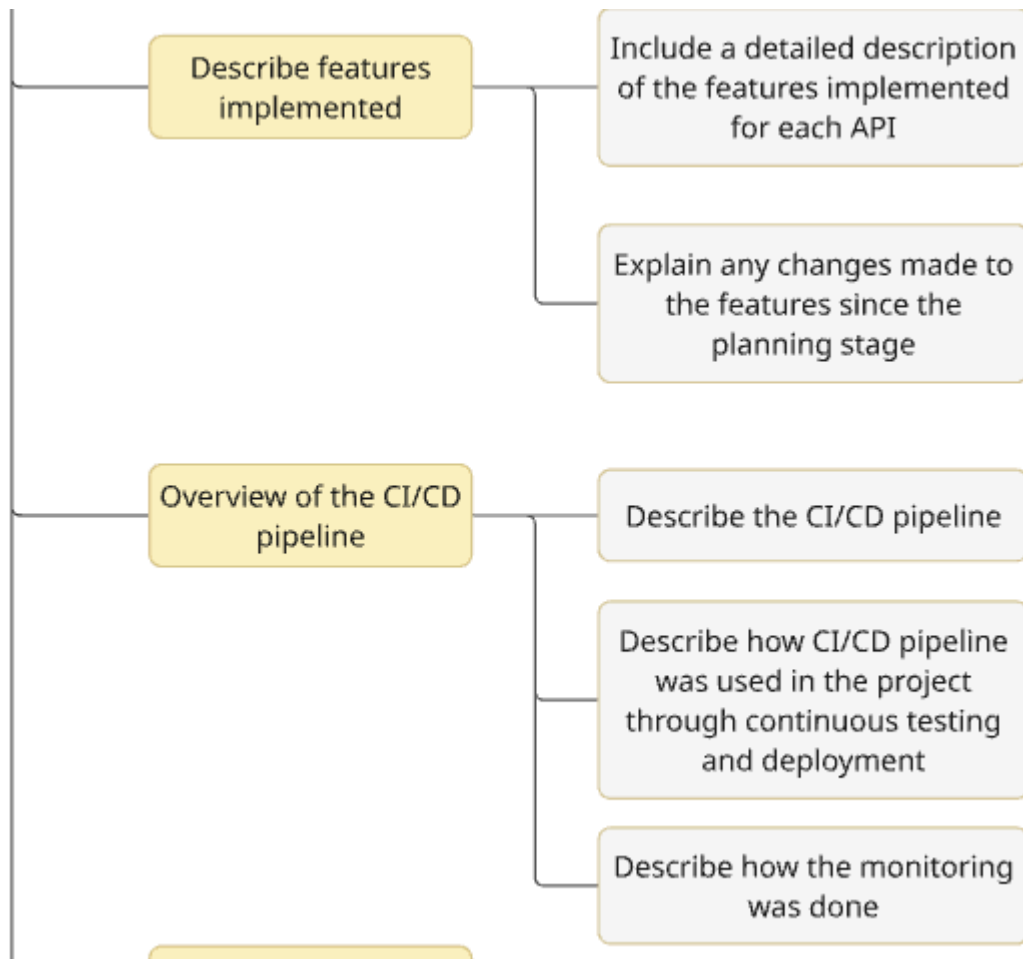


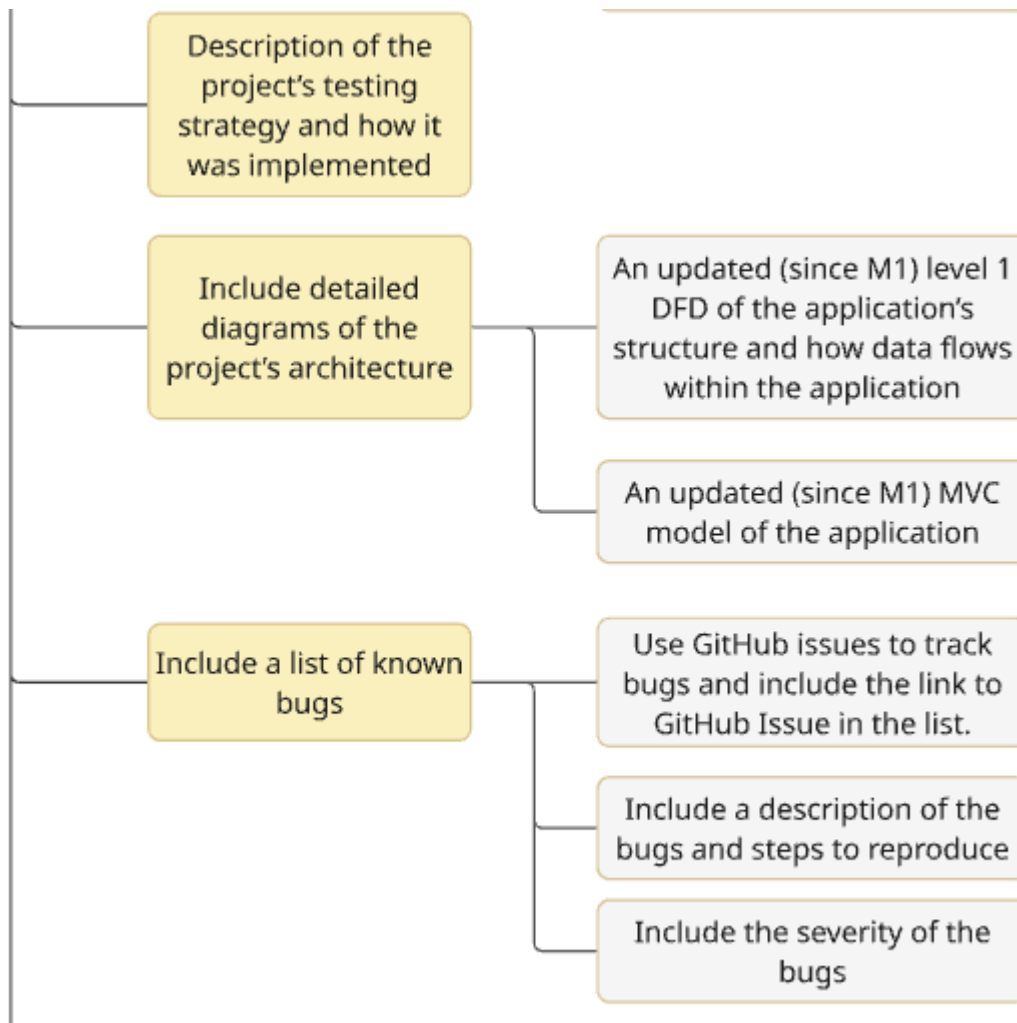
Peer Evaluation

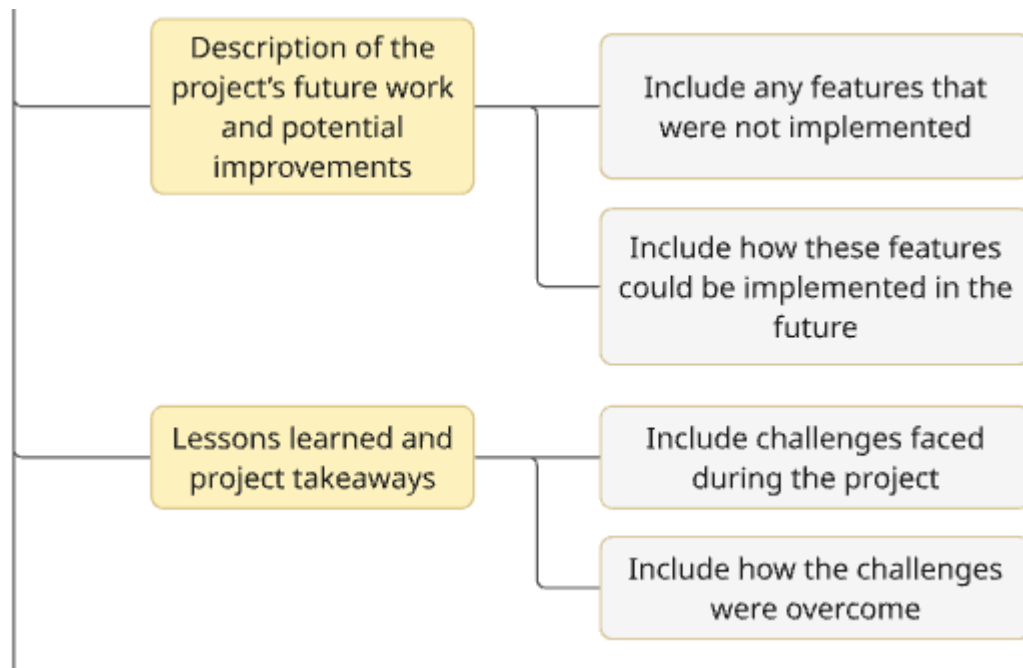
Submit a peer
evaluation using link

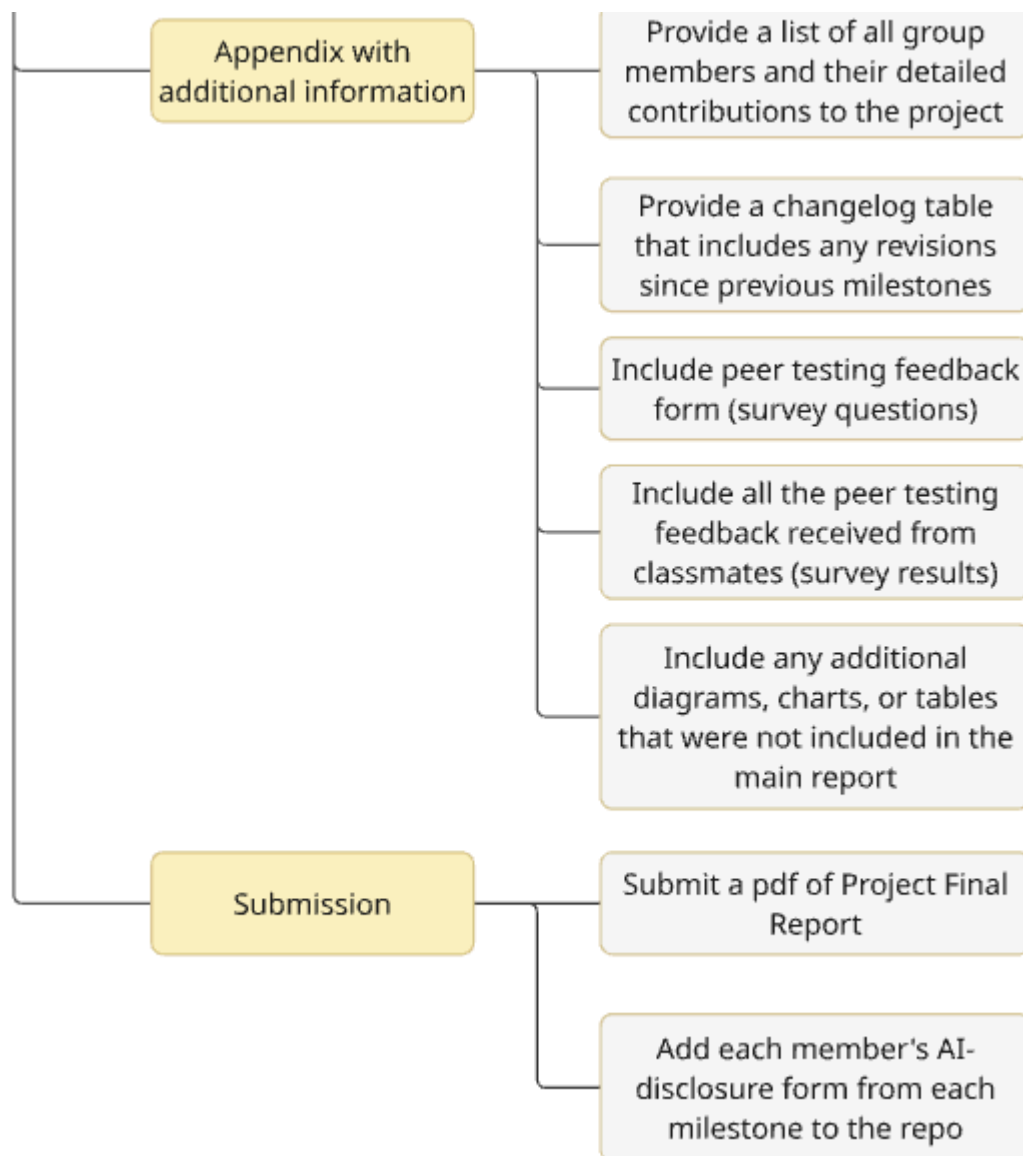
Submit the pdf on
canvas (no AI form)

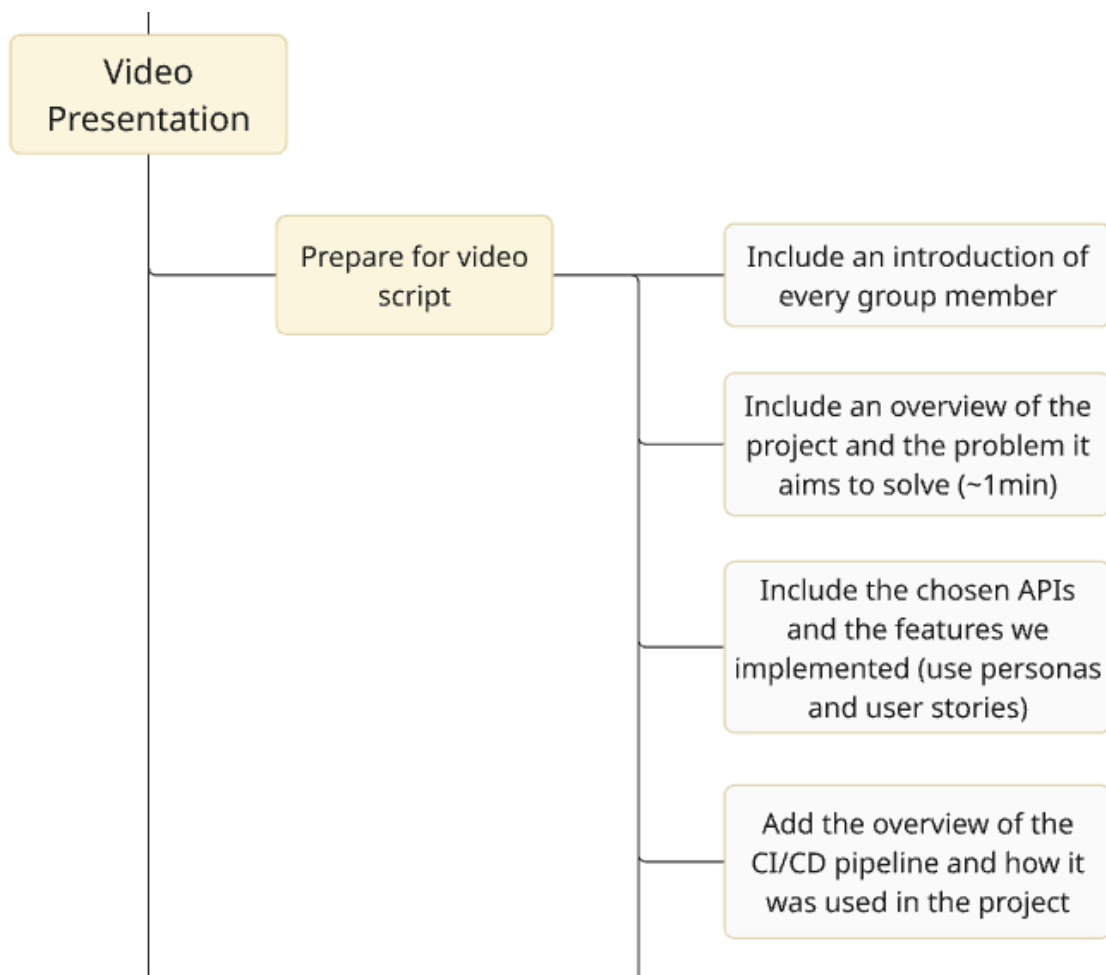


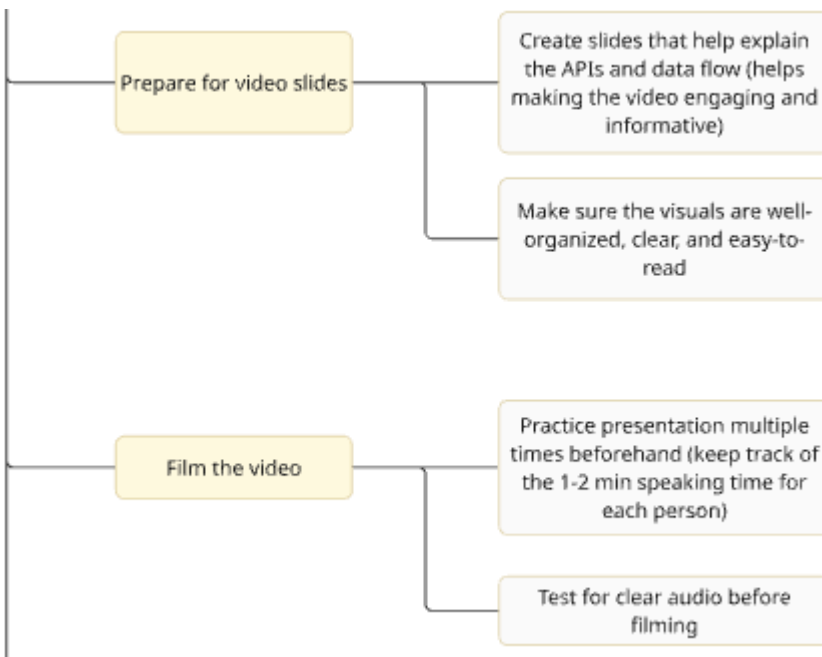
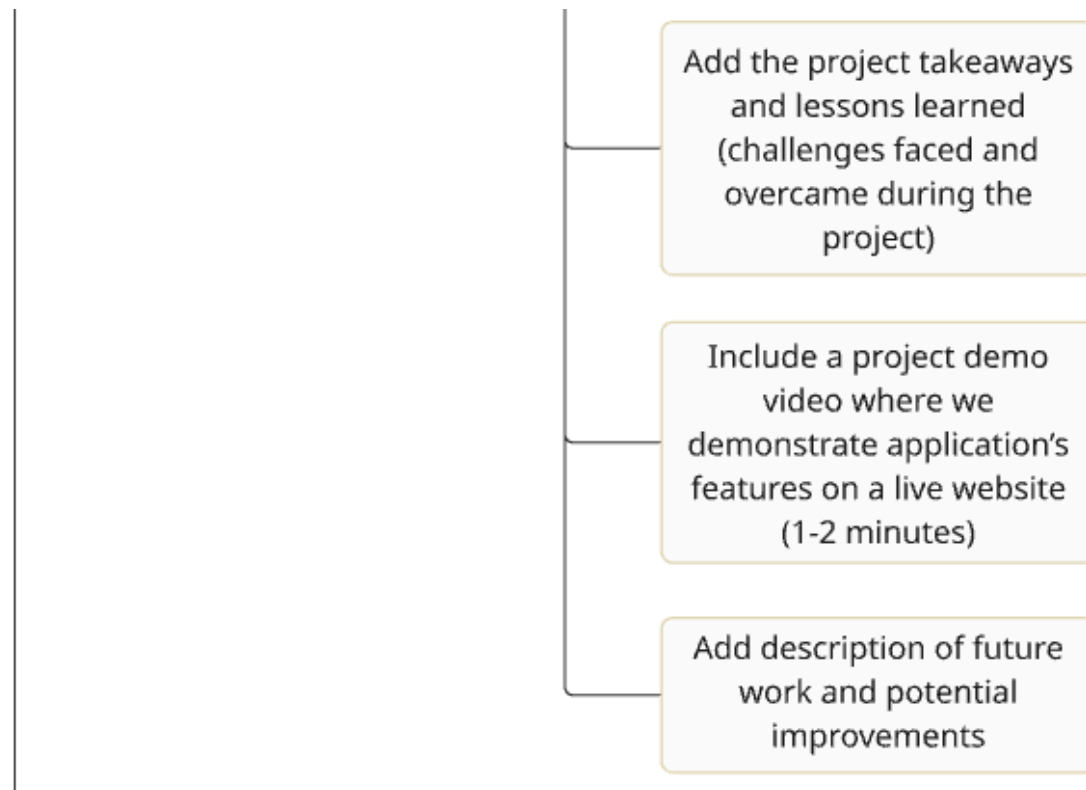


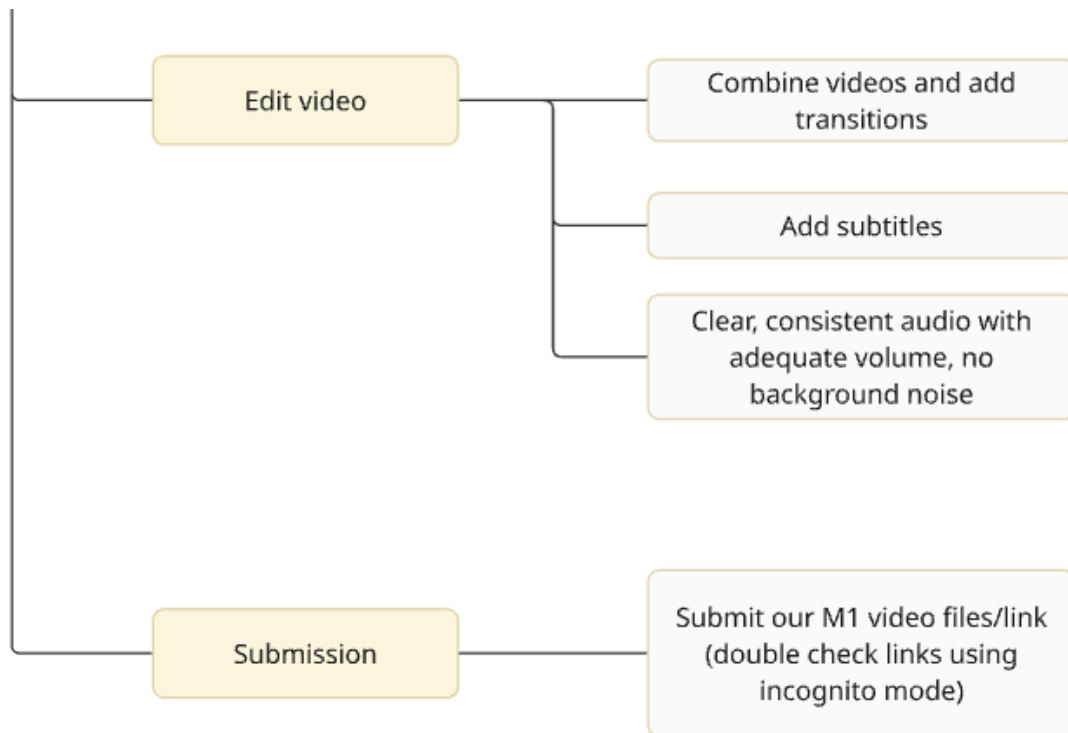












6. Project Schedule (Google Sheets)

Schedule Link:

https://docs.google.com/spreadsheets/d/1_TEz9PCmAOSPhJLkq0Gpt7I1nClpWL_Edy1DxFKhBCM/edit?usp=sharing

Schedule Screenshot:

PROJECT	Taiwan Explorers
GROUP	18 Flowers
PM + Team	Ryan Fu (PM) + (Alex Lee, Ikkyu Li, Tian Zheng)
FINAL PROJECT DUE DATE	August 8

Gantt Schedule colour code (Legend):

Red: High-Priority

- Tasks that must be completed first, or else they will affect the progress of other tasks

Orange: Mid-Priority

- Tasks that are similar to red colour tasks in terms of importance, but will not affect the progress of other tasks

Green: Low-Priority

- Tasks that are fast/easy to do or that are not too important compared to other in-progress tasks

Gray: Done Early

- Tasks that are done days before the due date of the task

Cyan: Key milestone tasks based on WBS level 1 (Task titles with a star emoji at the end: ★)

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	ACTUAL FINISH DATE	DUE DATE
o	Milestone o				
o.1	Preparation of Milestone o ★	Group	5/23/25	6/9/25	6/9/25
o.1.1	Complete and sign group contract with TA approval	Group	5/23/25	5/23/25	6/9/25
o.1.2	Setup Project Repository	Alex	5/28/25	6/9/25	6/9/25
o.1.3	Research and find 4 APIs for the project	Group	5/26/25	6/4/25	6/13/25
o.2	Project Proposal Report ★	Group	5/26/25	6/13/25	6/13/25
o.2.1	A cover page with the project title, group members and link to github repository	Alex	5/29/25	6/2/25	6/13/25
o.2.2	Include an overview of the project	Group	5/26/25	6/3/25	6/13/25
o.2.3	Create 1 persona for each application user group	Ikkyu	6/4/25	6/6/25	6/13/25
o.2.4	List chosen APIs (4 total) and a brief description of each	Group	6/4/25	6/6/25	6/13/25
o.2.5	Include a brief description of the features	Group	6/4/25	6/6/25	6/13/25
o.2.6	Include a low-fidelity storyboard	Group	6/7/25	6/13/25	6/13/25
o.2.7	Choose a front-end technology stack that this project will be based on	Tian	5/28/25	5/30/25	6/13/25
o.2.8	Submission of Milestone o	Ryan	6/13/25	6/13/25	6/13/25

1	Milestone 1				
1.1	Project Planning Report ★	Group	6/18/25	7/4/25	7/4/25
1.1.1	Finalize 2 APIs we will be using	Alex	6/29/25	6/30/25	7/4/25
1.1.2	Determine 3 features for each API (6 total)	Alex, Ryan	6/29/25	7/1/25	7/4/25
1.1.3	Convert the storyboard to mid-fidelity	Alex, Tian	6/29/25	7/4/25	7/4/25
1.1.4	Choose a SDLC model that we will follow	Tian	6/25/25	6/30/25	7/4/25
1.1.5	Develop a work breakdown structure (WBS)	Ikkyu	6/27/25	6/30/25	7/4/25
1.1.6	Project schedule with milestones and deadlines	Ryan	6/27/25	7/3/25	7/4/25
1.1.7	Risk assessment of potential issues with mitigation strategies	Alex	6/29/25	6/30/25	7/4/25
1.1.8	Data Flow Diagrams (level 0 and level 1)	Ikkyu	6/30/25	7/2/25	7/4/25
1.1.9	MVC model diagram	Tian	7/1/25	7/4/25	7/4/25
1.1.10	Appendix with additional information	Ryan	7/1/25	7/3/25	7/4/25
1.1.11	Submission of Project Planning Report	Ryan	7/4/25	7/4/25	7/4/25
1.2	Video Presentation ★	Group	7/4/25	Not Started	7/7/25
1.2.1	Prepare for video script	Group	7/3/25	Not Started	7/7/25
1.2.2	Prepare for video slides	Alex, Tian	7/3/25	Not Started	7/7/25
1.2.3	Film the video (6 min max)	Group	7/4/25	Not Started	7/7/25
1.2.4	Edit the video	Alex	7/4/25	Not Started	7/7/25
1.2.5	Submission of video presentation	Ryan	7/7/25	Not Started	7/7/25
1.3	Peer Review ★	Group	7/7/25	Not Started	7/18/25
1.3.1	Fill out the peer review form	Group	7/7/25	Not Started	7/18/25

2	Milestone 1.5				
2.1	Progress Report ★	Group	7/18/25	Not Started	Week of July 21-25
2.1.1	Complete expected progress and write a 1-page progress report	Group	7/18/25	Not Started	Week of July 21-25
2.1.2	Submission of Progress Report	Ryan	7/18/25	Not Started	Week of July 21-25
2.2	Check-in Meeting ★	Group	After Progress Report	Not Started	Week of July 21-25
2.2.1	Sign up and select a time slot for the check-in meeting	Ryan	After Progress Report	Not Started	Week of July 21-25
2.2.2	Attend the check-in to receive feedback on the progress of our project (has late-penalty)	Group	After Progress Report	Not Started	Week of July 21-25

3	Milestone 2				
3.2	Peer Evaluation ★	Group	7/28/25	Not Started	8/4/25
3.2.1	Submit a peer evaluation using link	Group	7/28/25	Not Started	8/4/25
3.2.2	Submit the pdf on canvas (no AI form)	Group	7/28/25	Not Started	8/4/25
3.1	Source Code ★	Group	7/8/25	Not Started	8/6/25
3.1.1	Ensure a functional website with completed features	Ikkyu	7/23/25	Not Started	8/6/25
3.1.2	Application testing	Tian	7/25/25	Not Started	8/6/25
3.1.3	Fulfill 10 Usability Heuristics in the application	Tian	7/30/25	Not Started	8/6/25
3.1.4	Ensure no errors, console.logs, or warnings in the Project Console	Ikkyu	7/30/25	Not Started	8/6/25
3.1.5	Complete root project README file	Alex	7/31/25	Not Started	8/6/25
3.1.6	Push source code to repository	Group	8/2/25	Not Started	8/6/25
3.1.7	Submission of Source Code	Ryan	8/6/25	Not Started	8/6/25
3.3	Project Final Report ★	Group	7/30/25	Not Started	8/6/25
3.3.1	Cover Page with title, members, links, video and site	Alex	7/30/25	Not Started	8/6/25
3.3.2	Analysis of user needs	Ryan	7/30/25	Not Started	8/6/25

3.3.3	Analysis of SDLC model	Tian	7/30/25	Not Started	8/6/25
3.3.4	Describe features implemented	Ikkyu	7/30/25	Not Started	8/6/25
3.3.5	Overview of the CI/CD pipeline	Alex	8/1/25	Not Started	8/6/25
3.3.6	Description of the project's testing strategy and how it was implemented	Tian	8/1/25	Not Started	8/6/25
3.3.7	Include detailed diagrams of the project's architecture	Ryan	8/1/25	Not Started	8/6/25
3.3.8	Include a list of known bugs	Ryan	8/2/25	Not Started	8/6/25
3.3.9	Description of the project's future work and potential improvements	Ikkyu	8/1/25	Not Started	8/6/25
3.3.10	Lessons learned and project takeaways	Group	8/2/25	Not Started	8/6/25
3.3.11	Appendix with additional information	Ryan	8/4/25	Not Started	8/6/25
3.3.12	Submission of Project Final Report	Ryan	8/6/25	Not Started	8/6/25
3.4	Video Presentation ★	Group	8/5/25	Not Started	8/8/25
3.4.1	Prepare for video script	Ryan	8/5/25	Not Started	8/8/25
3.4.3	Film the video	Group	8/7/25	Not Started	8/8/25
3.4.4	Edit video	Alex	8/7/25	Not Started	8/8/25
3.4.5	Submission of video presentation	Ryan	8/8/25	Not Started	8/8/25

7. Risk Assessment

Low-risk (5 risks)

Low-risk 1

- **Description:** User interface (UI) issues that don't abide by usability heuristics
- **Mitigation:** Regularly perform analytical evaluation with frameworks such as Nielsen's 10 to ensure the UI is user-friendly.

Low-risk 2

- **Description:** Typos or temporary elements left in the UI

- **Mitigation:** Schedule group meetings to review the UI before submissions.

Low-risk 3

- **Description:** Group heavily delays communication
- **Mitigation:** Set mandatory check-ins with shared task boards and reminders every week multiple times.

Low-risk 4

- **Description:** Wasting time on useless or redundant features
- **Mitigation:** Confirm priorities with a group or TA and stick to a reasonable feature list.

Low-risk 5

- **Description:** Forgetting to push/pull or using improper/unclear commit messages
 - **Mitigation:** Assign a manager who enforces clear version control rules and conventions to abide by.
-

Medium-risk (5 risks)

Medium-risk 1

- **Description:** Inaccurate scheduling and time estimates for tasks
- **Mitigation:** Use Agile sprints and track task durations to improve future estimates.

Medium-risk 2

- **Description:** Scope-creep gets too large
- **Mitigation:** Stick to a Minimum Viable Product (MVP) and document all added features with rationale of everyone's circumstances (technological knowledge and time).

Medium-risk 3

- **Description:** Team member unavailable for a period of time
- **Mitigation:** Have flexible task assignments and strong communication ahead of time.

Medium-risk 4

- **Description:** Group struggles at integrating finalized APIs due to lack of experience
- **Mitigation:** Start API integration early, using mock data until APIs are set up and ready to be integrated.

Medium-risk 5

- **Description:** Misunderstanding user requirements and needs
 - **Mitigation:** Use user stories, mockups, and perspectives of real people in the first third of the project.
-

High-risk (5 risks)

High-risk 1

- **Description:** Group member becomes unavailable indefinitely or refuses to contribute
- **Mitigation:** Assign roles/tasks to people with overlapping knowledge and communicate to the TAs/instructor if necessary.

High-risk 2

- **Description:** Loss of substantial work without backup
- **Mitigation:** Use Git with frequent commits and backup critical files consistently.

High-risk 3

- **Description:** Major bugs close to the project deadline
- **Mitigation:** Don't ignore the warnings of "The Mythical Man Month" and allocate more time to the testing/debugging phase.

High-risk 4

- **Description:** Third-party services like APIs or libraries fail out of nowhere
- **Mitigation:** Have backup APIs or libraries planned in advance.

High-risk 5

- **Description:** Major team conflict that blocks collaboration

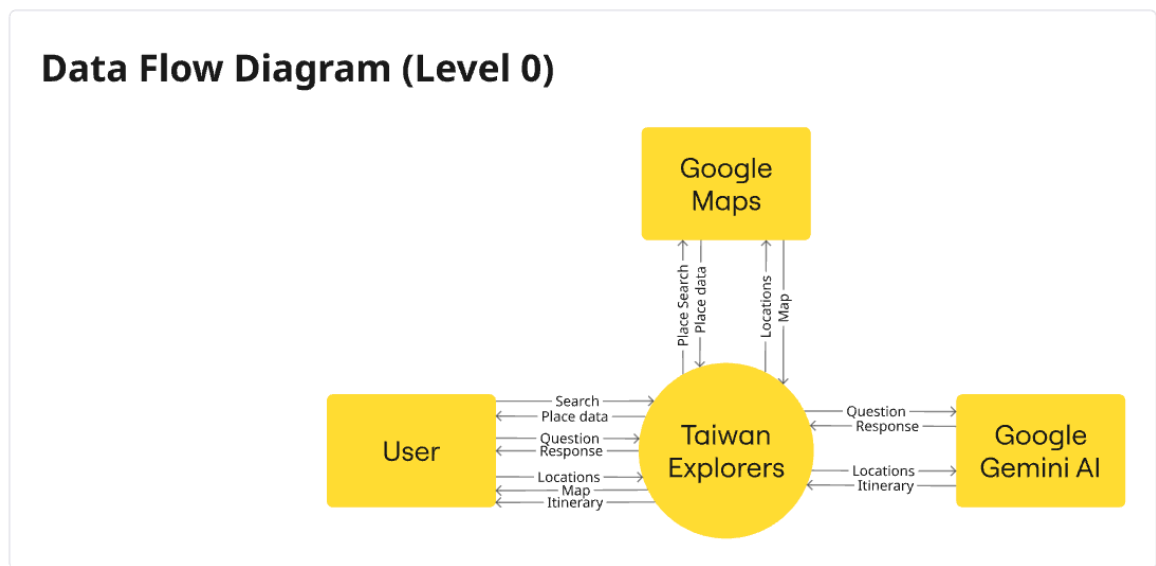
- **Mitigation:** Set conflict resolution guidelines and involve the PM and a neutral mediator early on.

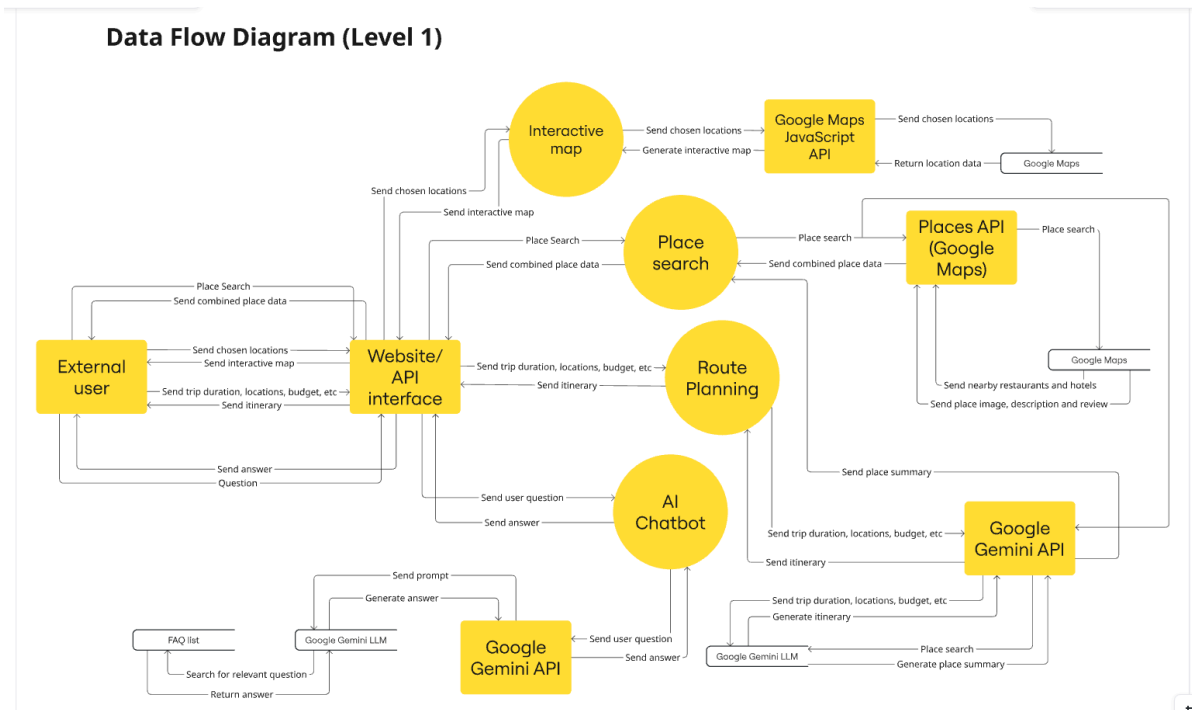
8. Data Flow Diagram Level 0 + 1 (Miro)

Miro Link:

https://miro.com/app/board/uXjVLM6JYXI=?share_link_id=135881325611

Data Flow Diagram Screenshot:





9. MVC Model (Miro)

Miro Link:

https://miro.com/app/board/uXjVLM6JYXI=?share_link_id=135881325611

Model

The model is contacted by the controller. According to received instructions, it accesses the API databases to fetch specific information and returns it back to the controller. The two API's that we will be using are Google Maps and Gemini. The Google Maps API will be used to display dynamic maps of various locations and the Gemini API will be used to help answer the user's questions.

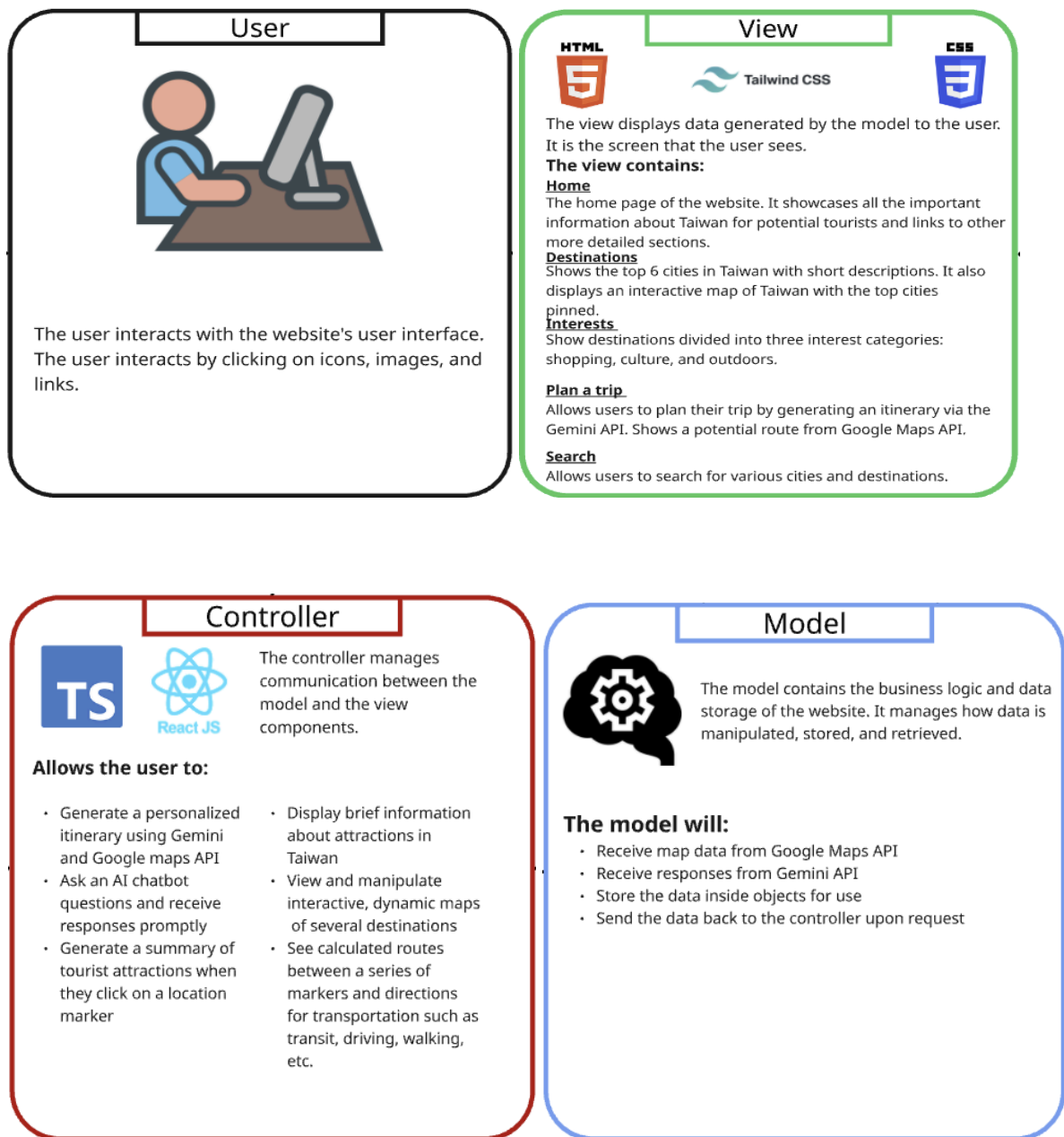
View

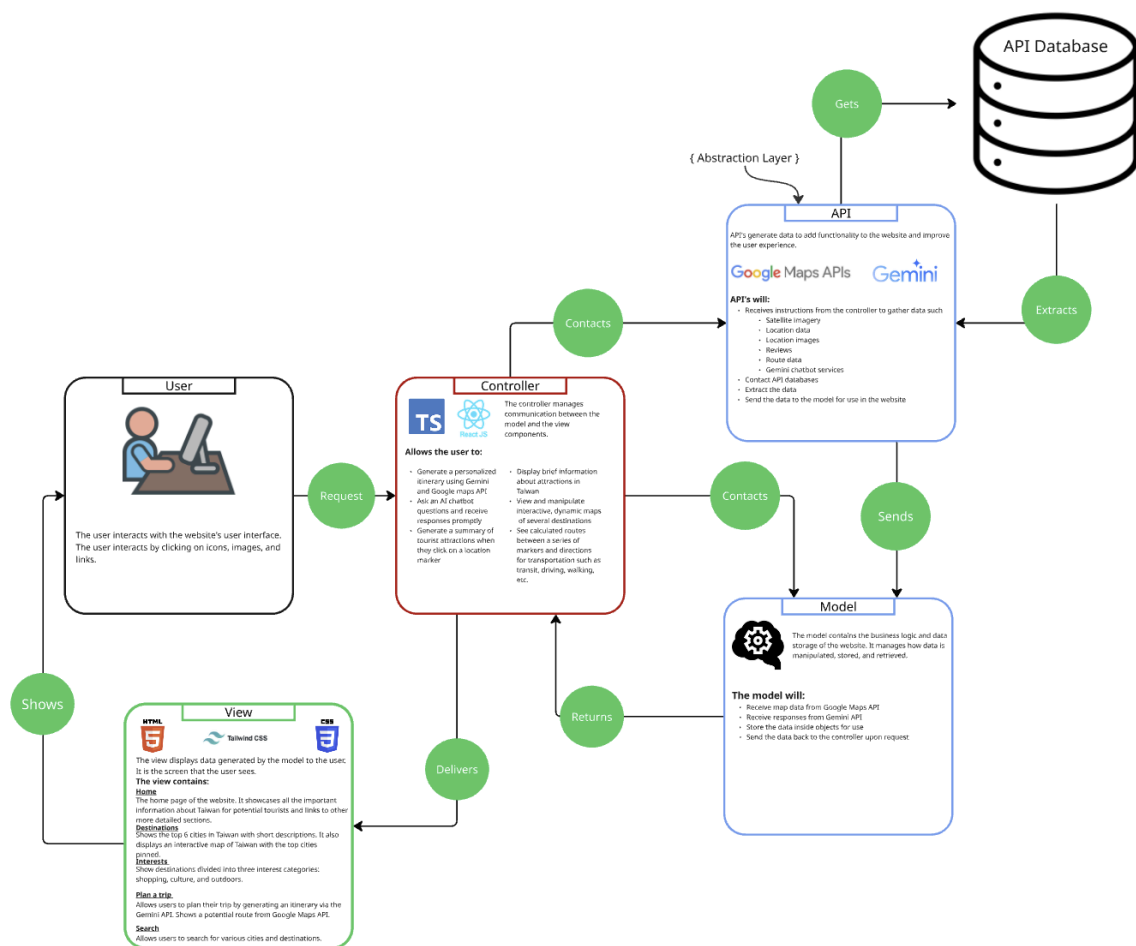
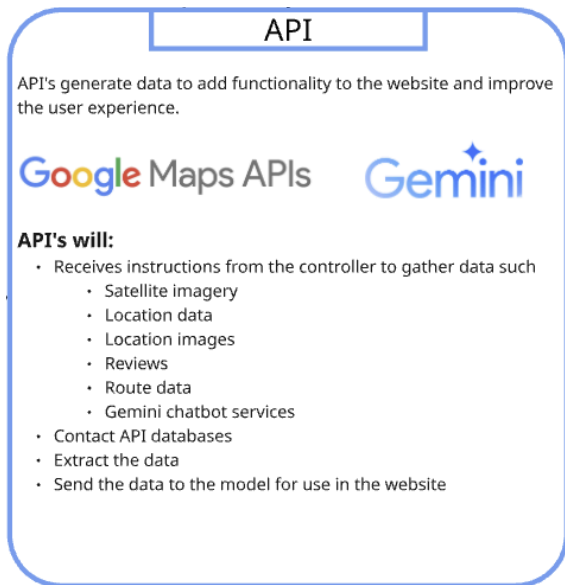
HTML and CSS are used to display content to users. These technologies make up the user interface of the application. When the controller is manipulated, changes to the UI are delivered to the view. Data that is extracted from API's is also delivered to the view via the controller. The view is shown to the user.

Controller

The controller receives requests from the user to perform certain actions. Based on those requests, it contacts the API's in the model section to execute them. The controller will then receive data extracted from the model and deliver them to the view to be shown to the user. The controller's logic will be developed with JavaScript and react so that it can properly handle various user requests.

MVC Model Screenshot (Individual Components + Full View are included):





10. Appendix

1. **Provide a list of all group members and their detailed contributions to the project**

Alex Lee

Contributions to the Task:

- Milestone 0: Set up the project repository on GitHub, which included adding docs, src, and misc folders to the repository. Furthermore, wrote the cover page and created the initial layout for the Project Proposal Report, including the project title, group member names, and GitHub repository link.
- Milestone 1: Finalized the 2 APIs and identified three features for each API, ensuring that each of the features is within the scope of the project by including three benefits for each feature. Transformed the low-fidelity storyboard into a mid-fidelity prototype using Figma. Furthermore, completed the risk assessment by identifying 15 unique risks categorized by severity as low, medium, and high.

Ikkyu Li

Contributions to the Task

- Milestone 0: Created detailed user personas to represent different types of travellers, including a student, elderly, explorer, and hiker, by outlining their needs, tech experience, and pain points. Furthermore, she wrote the corresponding user stories to capture their goals and expectations when using the application.
- Milestone 1: Constructed the Work Breakdown Structure (WBS) to visualize the tasks and responsibilities across milestones. Furthermore, worked on and completed both level 0 and level 1 of the Data Flow Diagrams (DFDs)

Ryan Fu

Contributions to the Task

- Milestone 0: In charge of the submission of Milestone 0, which involved reviewing all the completed tasks, checking that each required file was present and properly formatted, and making sure everything met the project requirements, which ensured that all files were complete and submitted on time.
- Milestone 1: Created and completed the project schedule with start/end dates and due dates for all tasks based on the WBS milestone breakdown. Assigned and accomplished the Appendix With Additional Information, which included a changelog and group contributions. Furthermore, handled the final submission

of the planning report for Milestone 1, while ensuring that all files were correctly submitted.

Tian Zheng

Contributions to the Task

- Milestone 0: Wrote the explanation for the tech stack we chose as a group which included HTML, CSS, Tailwind, JavaScript, and React. Furthermore, the explanations described how each tool will be used in the project, focusing on things like making the site responsive and organized.
- Milestone 1: Researched and selected the Agile Kanban SDLC model as the final project's development approach, which includes a detailed explanation of why it was a good fit, highlighting its flexibility, teamwork, and how it helps track tasks using GitHub. Furthermore, worked on and completed the MVC model diagram to illustrate how the project is organized and how data flows between its various components, including the user interface, logic, and database.

Group Contributions

Contributions to the Task as an entire group

- Milestone 0: Worked together to answer the questions and sign the group contract, research and describe the APIs and their possible features (with each member being assigned on researching and describing one API), decided on the front-end tech stack to use for the project as a group, and create the low-fidelity storyboard using Figma.
- Milestone 1: Worked together and checked in with each other through weekly meetings. Additionally, each member supported others by giving reviews and feedback on their tasks to ensure everything was complete and free of issues.

2. Provide a changelog table that includes any revisions since the proposal

Date	Revisions made	Reason for revision
6/27/25	Added TypeScript as a tech stack that will be used for the final project	Adding TypeScript to the tech stack improves reliability, team collaboration, and long-term maintainability. Furthermore, TypeScript allows us to declare data types for variables we'll use for user input data, which aids us in detecting errors easily. These benefits will greatly support the progress and the success of this project/application.
7/1/25	Replaced the Street View feature in Google Maps API with a Route Calculation API	The original Google Maps API feature was replaced because the new feature is much more useful for helping users plan their trips and navigate around Taiwan.
7/1/25	Merged the Custom Budget Trip feature into the Itinerary Generator feature for Google Gemini API	These two features were merged into one because they had significant overlap between them. By combining them into one feature, it will simplify the user experience and help reduce the project scope for the team.
7/1/25	Added AI Summary of Places as a new Google Gemini API feature	Gave users quick summaries of locations to help users with research and decision-making.
7/1/25	Rewrote and renamed the AI Helper/FAQ feature	Improved clarity and alignment with the actual

	using Google Gemini API	functionality of the feature.
7/2/25	Refined the scope of the wireframe during the transition from low-fidelity storyboard to mid-fidelity prototype	To focus on essential pages and reduce visual clutter. Some features were simplified or removed to keep the prototype and application realistic, usable, and achievable within the project time limit.
7/3/25	Clarified and assigned group roles and responsibilities for Milestone 1 and all milestones onwards	Ensured accurate documentation of each member's work for submission and ensured every member has an equal amount of work.

3. Any additional diagrams, charts, or tables that were not included in the main report

N/A